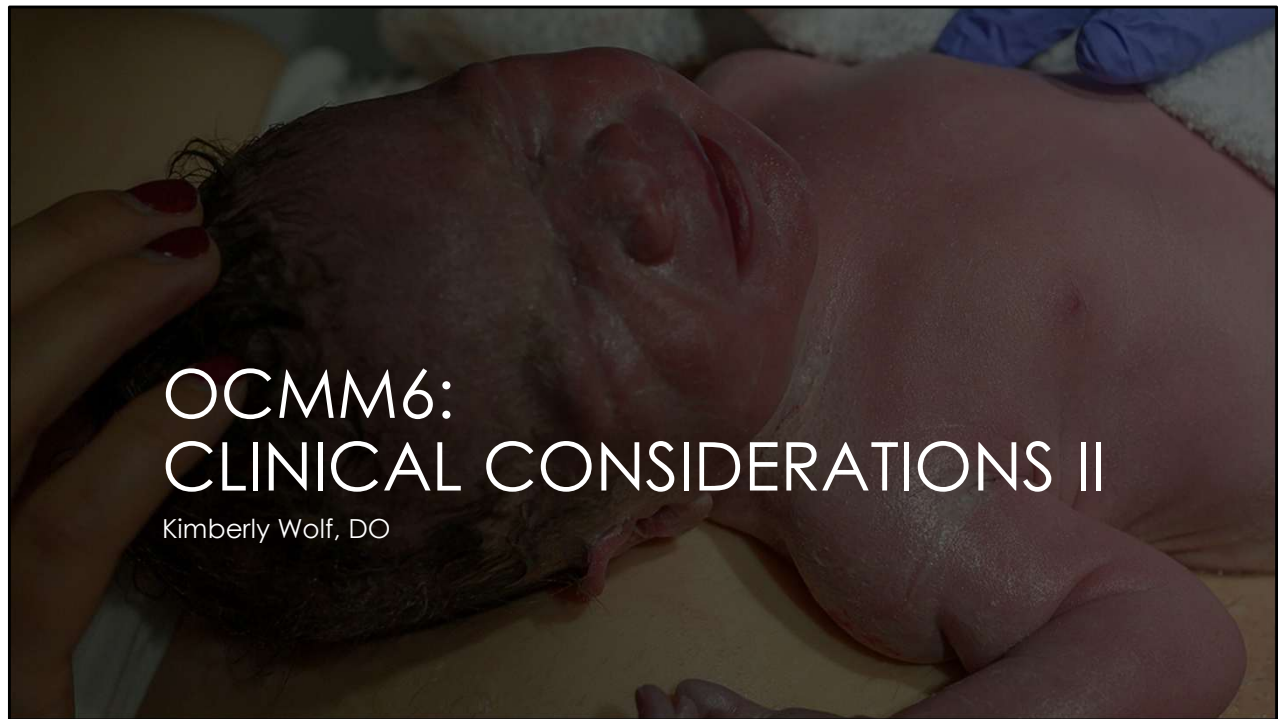




OCMM6: CLINICAL CONSIDERATIONS II

Kimberly Wolf, DO

Welcome to OCMM6: Clinical considerations part 2 lecture. This is one of my fave lectures because we'll be talking about the pediatric clinical considerations.

A photograph of a newborn baby lying down, possibly in a hospital bed. The baby is wearing a pink hospital gown and has a green pacifier in its mouth. The baby's head is tilted to the right, and there is a visible indentation on the left side of the head, suggesting plagiocephaly. The text "PLAGIOCEPHALY, TORTICOLLIS, AND FEEDING DYSFUNCTION" is overlaid in white capital letters on the image.

PLAGIOCEPHALY, TORTICOLLIS,
AND FEEDING DYSFUNCTION



OBJECTIVES

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:
 - 1. Plagiocephaly/torticollis and feeding dysfunction**
 2. Otitis Media
 3. Concussion
2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study
3. Compare and contrast normal motion and pathological motion of the individual cranial structures potentially associated with conditions discussed during the self-study.
4. Define relevant research as it relates to the conditions emphasized in the self-study.

CASE 1

- 2-month-old male infant presents for their well child check. Mom reports that the child is overall doing well, but that she has noticed the **child's head shape looks "funny"** to her. She has one other son who is 3 years old, and mom says his head never looked like this. Mom is also concerned that the **child always seems to look to the right**. She has tried multiple times to try to get him to look more to the left, especially while he's sleeping because she thinks that has something to do with his head shape changing. Mom **first noticed this change in head shape about a month ago**, but thinks it is getting worse. Mom reports that he is **fussy when doing tummy time so they don't do it every day**.
- Birth history: Born at 39 6/7 weeks via vaginal delivery. Mom was **in labor for 42 hours**, and then **pushed for 3 hours**. Ultimately, they used a **vacuum to assist** with delivery. Pregnancy was otherwise uncomplicated, and mom had good prenatal care from the first trimester. Mom and baby were both fine after delivery, and able to be discharged home together. Baby weight 8 lbs 6 oz at birth.
- Feeding history: mom has tried to **breastfeed** from day one, but has been very **difficult**. Baby had a **painful latch** for mom, and would really **only latch on to mom's right breast**. Mom has mostly been pumping and been giving her breastmilk to baby because it was so painful for her and frustrating for baby.

often
seen
with
torticollis

Let's start with a case...

full term is 40 weeks

CASE 1 CONTINUED

- ROS: funny-shaped head, right side preference, spits up occasionally, "colicky," feeding problems
- Meds: vitamin D
- Vaccinations: received Hep B at birth
- Allergies: NKDA
- PMHx/PSHx: no significant PMH, circumcision on DOL 1
- Trauma: none other than birth birth traumatic for all parties involved
- Family Hx: Mother and Father both alive and well with no major medical problems. One older brother who is also healthy.
- Social Hx: Lives with mom, dad, and brother. No daycare. No smoke in the home. One dog in the home.
- Developmental history: Able to briefly lift head while prone. Cooing. + social smile. Crossing midline with gaze.
on track developmentally
smiling in response to others

funny shaped head - other symptoms commonly associated

CASE 1

- VSS – growth appropriate
- PE:
 - Head: **Plagiocephaly with flattening of right occiput, slight fullness in right maxilla and frontal bones**, AFOSF, posterior fontanelle closed, **right ear anterior to the left**
 - EENT: PERRLA, EOMI, + red reflex, oropharynx clear, no apparent ankyloglossia, TMI, no bulging/fluid, nares patent
 - Neuro: normal tone, no focal deficits; + Moro, plantar/palmar grasp, Galant, rooting, and sucking, upgoing Babinski reflex
 - Abdomen: soft, NTND, no masses, no HSM, + BS x 4 quadrants
- OSE:
 - Cranium: **left lateral strain, right condylar compression, diminished PRM, externally rotated left temporal bone**
 - Cervical: rotated right throughout the cervical spine with slight hypertonicity of paraspinal musculature on the right, regionally sidebent left, **hypertonic L SCM**
 - Thoracic: **rotated right** throughout the thoracic spine with slight hypertonicity of paraspinal musculature on the right
 - Lumbar: **rotated right** throughout the lumbar spine with slight hypertonicity of paraspinal musculature on the right
 - Sacrum: diminished motion, S1 rotated right
 - Pelvis: restriction at the **right SI joint** with decreased mobility
 - UE: **elevated left clavicle**

pull inferior from scm pull mastoid medially and temporal into ER

PROBLEM LIST/DDX



"Funny shaped
head"

Positional Plagiocephaly
Craniosynostosis



Feeding
dysfunction

r/o ankyloglossia (tongue
tie) or other anatomic
abnormalities



Reflux

r/o anatomic
abnormalities such as
malrotation



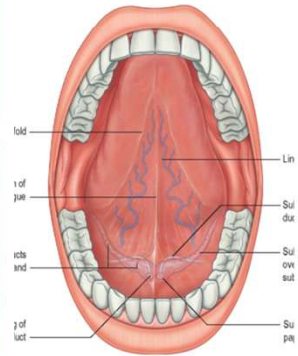
Colicky/fussy

Extensive differential

DIFFERENTIAL
INCLUDES:

- Cleft lip/palate
- Hypotonia
- Ankyloglossia
- Dysmorphic features – especially micrognathia
- Macroglossia
- Airway issues – malacias, stenosis, etc.

stridorous resp



small underdeveloped jaw

PLAGIOCEPHALY/TORTICOLLIS

- Relevant Anatomy:
 - Newborn skull
 - Occiput **post aspect of head**
 - SBS – lateral strain
 - Condyles
 - Jugular foramen
 - CN 9-11
 - Temporal bone
 - Direct attachment with SCM
 - TMJ articulation
 - Tongue muscular attachments
 - Hyoid
 - Sternocleidomastoid
 - Sternum
 - Clavicle
 - Mastoid process of temporal bone



Newborn skull – pics ahead but seen some of these skulls in labs. Flexible and malleable because of their sutures. 30 separate bones at birth but many of those are in several pieces – occiput, temporal, sphenoid so many places where restrictions can happen where they wouldn't in an adult skull

Lateral strain – MC in plagio – will review this in upcoming lecture

Condyles – jugular foramen but also hypoglossal canal so influencing CN 12 as well

Temporal – direct SCM attachment, TMJ articulation influences feeding mechanics, tongue attached to hyoid and mandible. Hyoid extends directly from temporal bone in infants and tucked underneath mandible. Hyoid with several muscular attachments including omohyoid to scapula, sternum, etc

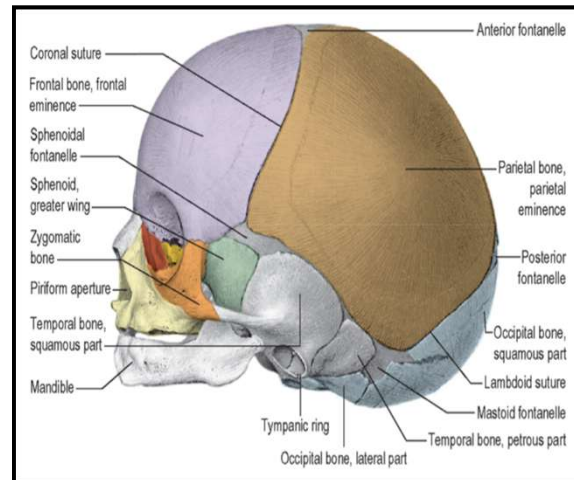
SCM – explains why this child's clavicle likely elevated on exam. Mastoid process of temporal bone – leads to externally rotated temporal bone on the left. Pulls medially on the mastoid process which moves the bone into ER

CRANIAL ANATOMIC CONSIDERATIONS

Neurocranium – 9 bones

- Occipital (4 parts at birth)
- 2 parietals
- 2 Frontals (2 parts at birth)
- 2 temporals (3 parts at birth)
- Sphenoid (3 parts at birth)
- Ethmoid

mandible in 2 and maxilla in 2 pieces

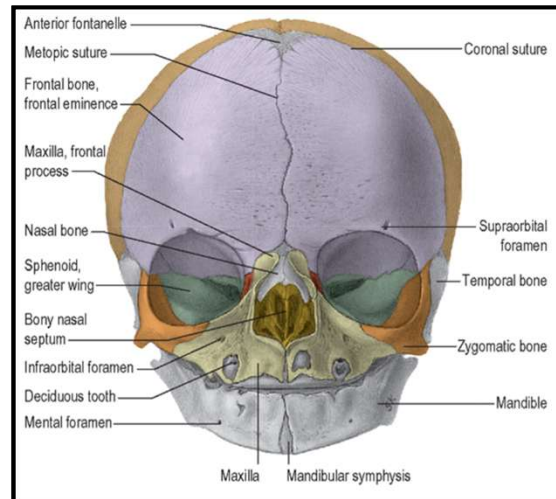


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CRANIAL ANATOMIC CONSIDERATIONS

Viscerocranium/facial skeleton

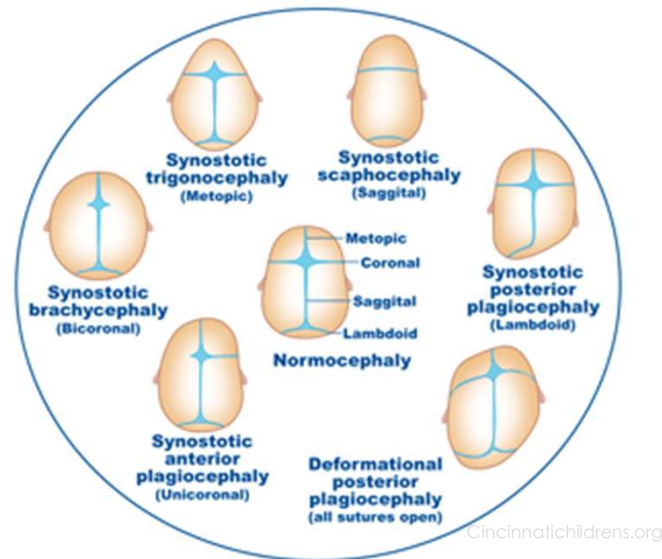
- 13 bones
 - 3 nasal bones
 - 2 maxillae
 - 2 lacrimal bones
 - 2 zygoma
 - 2 palatine
 - 2 inferior nasal conchae
 - Vomer
- Other
 - Mandible
 - Hyoid
 - Middle Ear Ossicles



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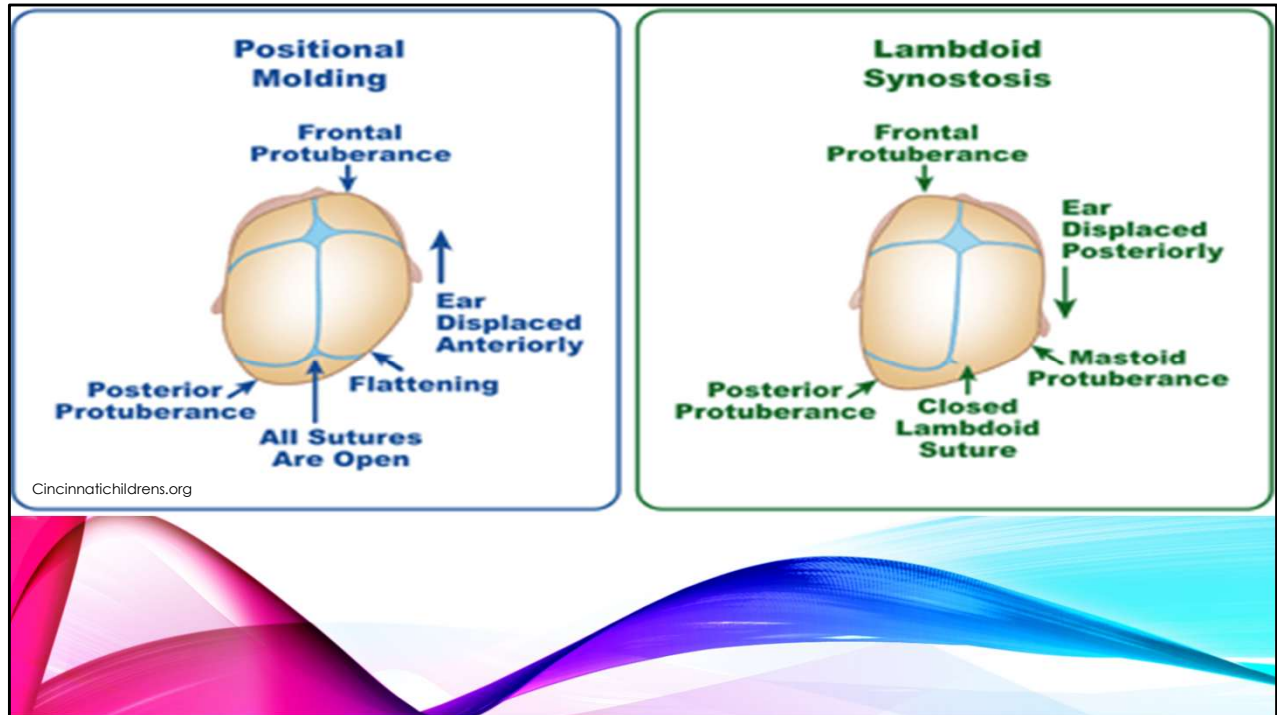
CRANIOSYNOSTOSIS

- Premature fusion of 1+ sutures
- 1: 2000-2500 births worldwide
 - Increased in uterine deformities and multiples, warfarin/valproate use, hyperthyroidism
- Restricted perpendicular growth leads to compensatory parallel growth
- Predictable deformities based on which suture fused
 - Sagittal 56%, Coronal 25%; multiple 13%



When look at plagio need to make sure that you don't miss craniosynostosis
Note the difference between deformational posterior plagiocephaly and synostotic posterior plagiocephaly

growing everywhere else - ear will stay put and get compensatory growth on opposite side



A closer look at these differences

Not ear positioning and frontal prominence as main differences

surgical condition - need to refer to neurosurgery

GENETICS

- Most single suture craniosynostosis is spontaneous
- Multiple sutures usually part of genetic syndrome
 - Mutations seen in fibroblast growth factor receptor family (FGFR)
 - Most are single gene mutations
 - Apert, Crouzon, Pfeiffer syndromes

Apert Syndrome



Crouzon Syndrome



POSITIONAL PLAGIOCEPHALY

- Aka deformational plagiocephaly
- Risk factors:
 - Torticollis or other conditions that decrease ROM
 - Supine sleep position
 - Decreased activity level
- Males > Females
- Risk for flattening at birth:
 - Multiples
 - Cephalohematoma
 - Primiparity
 - Long labor
 - Assisted deliveries



www.aqpublications.org

Back to sleep campaign – increased risk of SIDS/SUIDS- ABCs
Minimal tummy time just like in our case
Long labor and assisted delivery just like our case as well

OSTEOPATHIC CONSIDERATIONS FOR THE NEWBORN



- Labor forces + anatomy → compression
- Interplay between PRM and thoracic respiration
 - Usually, primary (cranial) respiration drives secondary (thoracic) inhalation
 - First breath influences restoration of normal cranial motion
 - Crying and suckling assist
 - Compression followed by expansion that occurs in natural birth “resets” the autonomics (similar to CV4 technique)

Image: <http://dou-la-la.blogspot.com/2010/08/postpartum-ocd-part-2-of-2-mom-who.html>

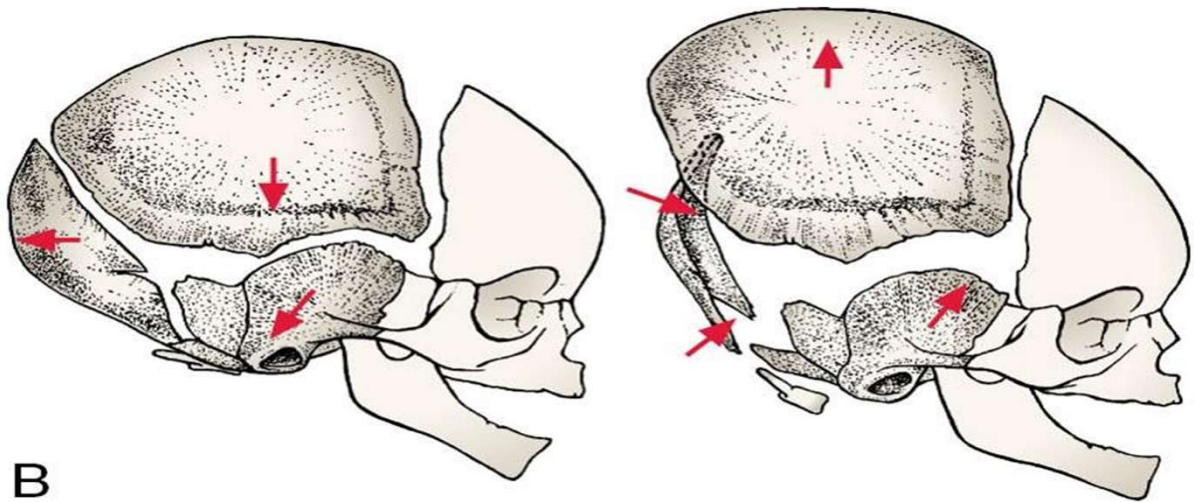
Can be altered by C-section or vacuum assisted deliveries

Mom had 3 hours of pushing and 42 hours of labor

First breath – should be a loud cry to help

Molding seen here – “normal” but does it look comfy to you?

CRANIAL MOLDING AFTER VAGINAL BIRTH



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<https://classconnection.s3.amazonaws.com/372/flashcards/2357372/jpg/molding-144A3DDC6686D92DF5A.jpg>

should resolve within min to hours for it to be normal molding



RESEARCH

- **JOM - Relation of disturbances of craniosacral mechanisms to symptomatology of the newborn: study of 1250 infants**
 - Goal: Explore relationship between anatomic-physiologic disturbances of the craniosacral mechanism and symptoms
 - 1250 babies assessed
 - <12% had "normal" cranial mechanisms (including 6 that had a single area of restriction likely to resolve spontaneously with only breathing, crying, and sucking).
- **JOM – Osteopathic Evaluation of Somatic Dysfunction and Craniosacral Strain Pattern Among Preterm and Term Newborns**
 - Occipital bone presented with highest rate of intraosseous lesions
 - Highest rate of SD – pelvis



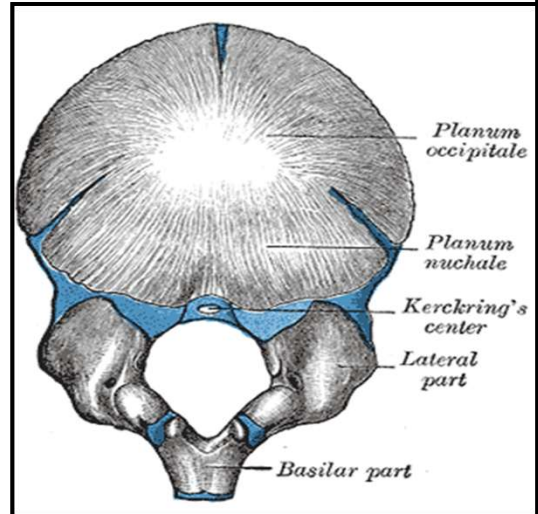
Frymann, V. "Relation of disturbances of craniosacral mechanisms to symptomatology of the newborn: study of 1250 infants." *JAOA*, vol. 65, pp. 1059-1075, June 1966. Reprinted in *Clinical Cranial Osteopathy*. Ed. Richard Feeley, D.O.
Image: <https://www.youtube.com/watch?v=DhSdYgskcRw>

One hundred fifty-five preterm and term newborns met the study's eligibility criteria. The highest rate of somatic dysfunction was found in the pelvic area of 63 newborns (40.7%). The sacroiliac joints were compressed unilaterally or bilaterally in 82 newborns (52.9%); the lumbosacral junction was restricted in 61 newborns (39.4%), and intraosseous lesions of the sacral bone were diagnosed in 57 newborns (36.8%). The spine accounted for somatic dysfunction in 38 newborns (24.5%), with the middle thoracic and lower thoracic areas restricted in 29 (18.7%) and 21 (16.8%) newborns, respectively. Sphenobasilar synchondrosis compression and lateral-vertical strain were diagnosed in 57 newborns (36.8%), with the sagittal and the coronal sutures found restricted in 35 (22.6%) and 30 (19.4%) newborns, respectively. The occipital bone presented the highest rate of intraosseous lesions, with the left condyle compressed in 48 newborns (31%), the right condyle in 46 newborns (29.7%), and the squama in 38 newborns (24.5%).

THE OCCIPUT

A-P at internal surface

- 4 parts at birth
- **Hypoglossal canal forms between the occipital condyles**
 - CN XII
- **Jugular foramen forms between temporal and occiput**
 - CN IX, X, XI
- **Cartilage easily compressible**
 - Birth trauma or intrauterine
 - Swelling can lead to compression

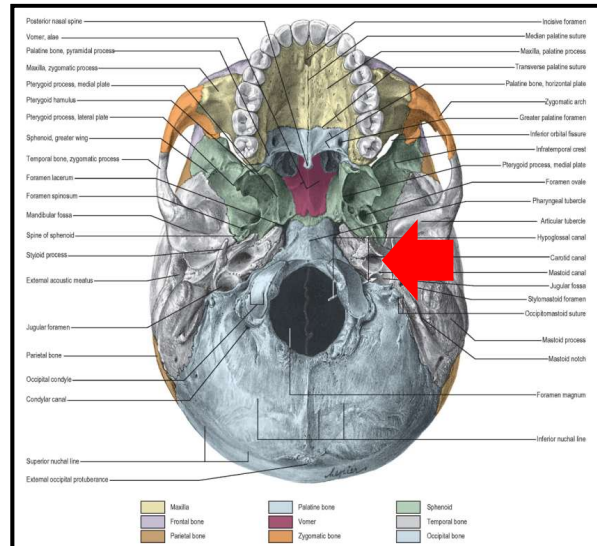


Gray's Anatomy Plate 131

Squamous portion is part of vault so allows for accommodation – molding
Anywhere you see blue is cartilage at birth

GLOSSOPHARYNGEAL NERVE IX

- Branchial motor – stylopharyngeus muscle (elevates pharynx to allow **swallowing**)
- Visceral motor – parasympathetic of parotids
- Visceral sensory – from carotid sinus and body
- General sensory – from TM, **upper pharynx, posterior 1/3 of the tongue**
- Visceral afferent – taste from posterior 1/3 of tongue
- Exits via the jugular foramen (**temporal/occiput**)

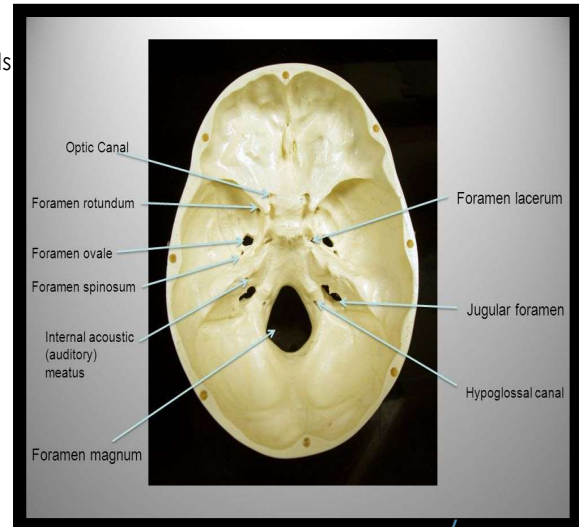


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ability to feed and swallow

VAGUS NERVE CN X

- General visceral efferent – parasympathetic to glands of mucus membranes of pharynx, larynx, organs in neck/thorax/abdomen
- Special visceral efferent – innervates skeletal muscles of pharynx/larynx
- General somatic afferent – sensation from EAM and TM
- General visceral afferent – from thoracic and abdominal viscera, aortic body and arch
- Special visceral afferent – taste of epiglottis region of tongue
- Exits via the jugular foramen



Reaches GE sphincter – reflux
Rest of gut - colicky

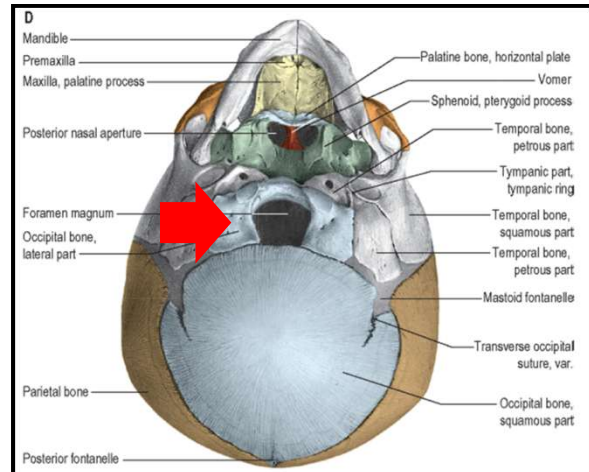
SPINAL ACCESSORY CN XI & HYPOGLOSSAL CN XII

○ Spinal Accessory CN 11

- Motor to **SCM and trapezius**
- SCM - bilateral to flex/extend head, unilaterally sidebends to same side & rotates to opposite side
- Exits via the jugular foramen

○ Hypoglossal CN 12

- Motor to **intrinsic tongue muscles**
- Exits via the hypoglossal canal

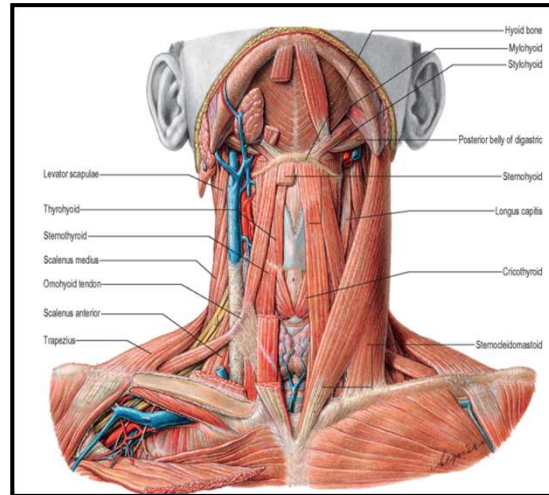


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HYOID & INTRINSIC TONGUE MUSCLES

- In infants, it extends straight from temporal bones, almost inside the mandible
- In adults, more inferior (anterior to C3)
- 6 muscles attach superiorly
 - Middle pharyngeal constrictor muscle, hyoglossus, digastric, stylohyoid, geniohyoid, mylohyoid
- 3 muscles attach inferiorly
 - Thyrohyoid, omohyoid, sternohyoid

including scap for omohyoid



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hyoid forms within membrane

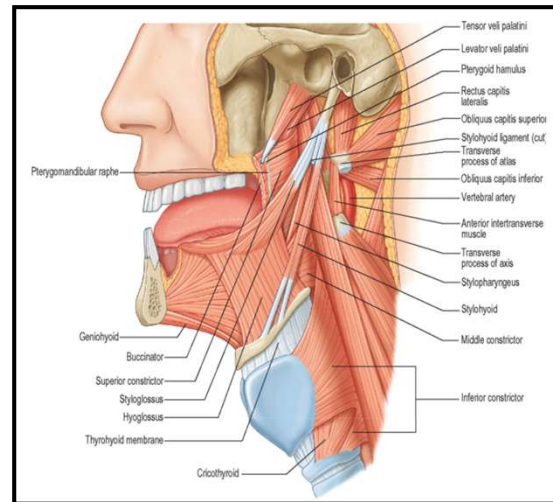
TONGUE MUSCLES

○ Extrinsic muscles

- Stabilize the tongue
- Allow for protrusion, retraction, side to side
- Genioglossus (mandible), hyoglossus, styloglossus, palatoglossus

○ Intrinsic muscles

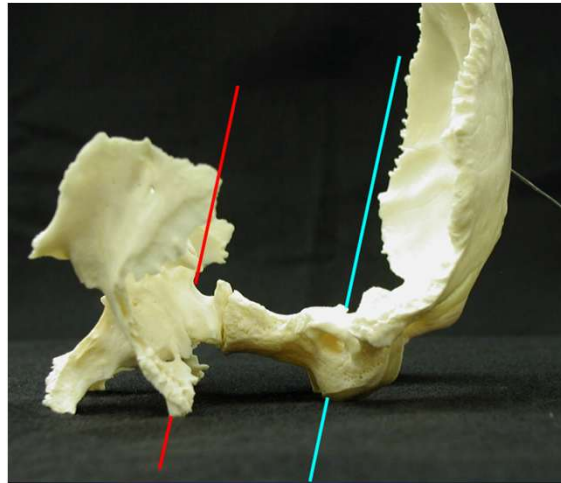
- Change shape, important for speech, swallowing, eating
- Lengthen/shorten, curling, flattening/rounding
- Superior longitudinal muscle, inferior longitudinal muscle, verticalis muscle, transversus muscle



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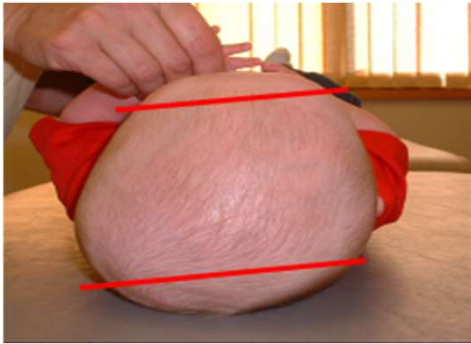
LATERAL STRAINS

- Named
 - Left or Right Lateral Strain
- Axes
 - 2 vertical axes
 - Body of the sphenoid
 - Through foramen magnum
- Rotation
 - Rotation in **same direction**
 - Shearing force at SBS causing Sphenoid and Occiput to rotate same direction on vertical axes
- Named
 - For position of base of sphenoid



LATERAL STRAINS

Left Lateral Strain

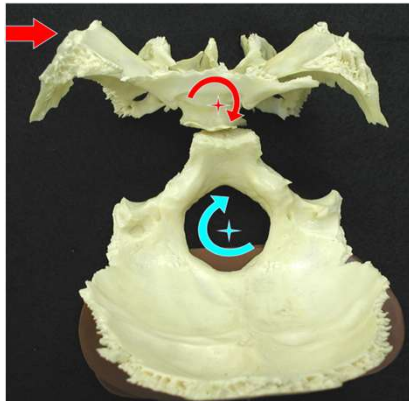


Right Lateral Strain

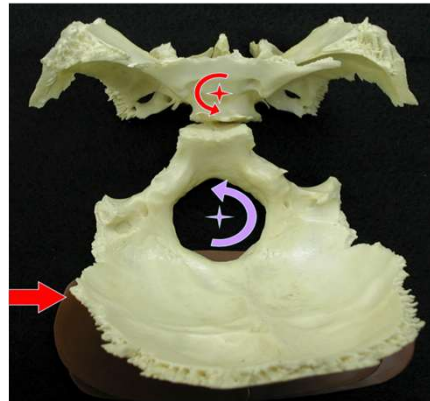


LATERAL STRAINS VIDEO

- http://www.youtube.com/watch?v=vZ1OOCC_DhI



Left Lateral Strain



Right Lateral Strain

LATERAL STRAINS AIR HANDS

Left Lateral Strain

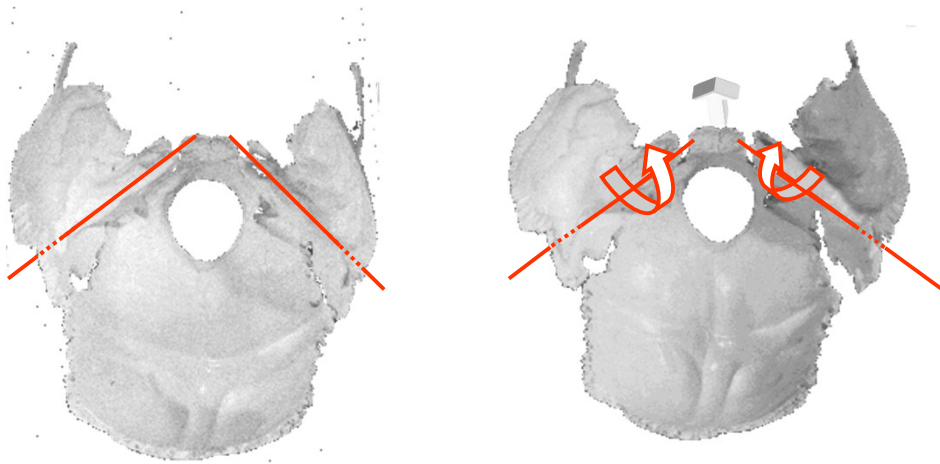


Right Lateral Strain



TEMPORAL BONE PRM MOTION

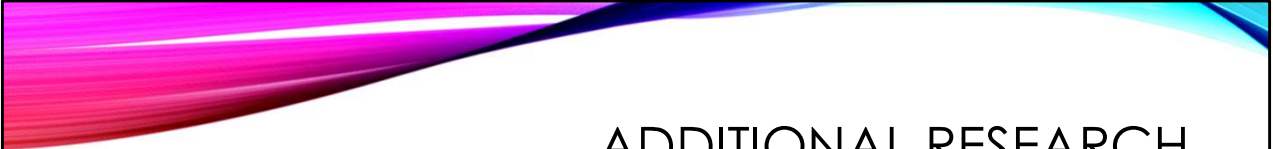
PAIRED BONE MOTION



important for scm

OSTEOPATHIC RESEARCH

- **AAO Journal – A Pilot Study: Osteopathic Treatment of Infants with Sucking Dysfunction**
 - Only 6 patients enrolled -<6 months
 - Compared pre-post feed fat estimations in breastmilk which improved to controls following OMT
- **JOM -Osteopathic Manipulative Treatment for the Treatment of Hospitalized Premature Infants With Nipple Feeding Dysfunction**
 - Case study of premature twins who avoided placement of G-tubes after OMT treatment helped their PO nipple feeding
- **Nationwide Children's Hospital Latch study - Pilot (in process)**
 - 11 patients enrolled
 - Improvement noted in OMT vs sham groups, but no statistical significance
 - Mothers whose infants received OMT did report less pain following OMT



ADDITIONAL RESEARCH

- **Use of OMT to Treat Congenital Torticollis and Positional Plagiocephaly in an Infant: A Case Report**
 - Case study on a 7-month-old – showed improvement, 3 treatments
- **Exploring the Impact of Osteopathic Treatment on Cranial Asymmetries Associated with Nonsynostotic Plagiocephaly in Infants**
 - *Complementary Therapies in Clinical Practice 2011*
 - 12 patients, 4 treatments – showed improvement
- **Osteopathic Center for Children**
 - Retrospective chart review showed improvement – working on prospective study



ADDITIONAL RESEARCH

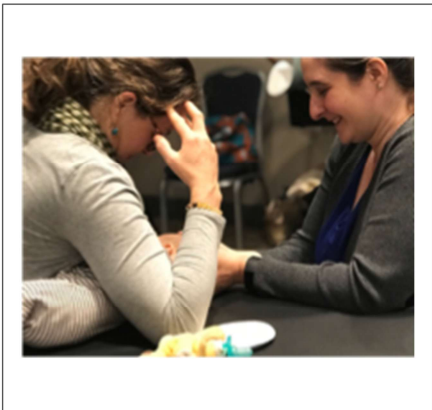
- **Effects of manual therapy on treatment duration and motor development in infants with severe nonsynostotic plagiocephaly: a randomised controlled pilot study**
 - 46 infants in a randomized controlled trial placed into standard treatment consisting of repositioning and orthotic treatment or to experimental group which included manual therapy
 - Treatment duration significantly shorter in experimental group compared to the control group
- **Infantile postural asymmetry and osteopathic treatment: a randomized therapeutic trial**
 - 32 infants with postural asymmetry randomly assigned to receive OMT or sham at 4 visits over course of 4 wks
 - Concluded OMT in first months of life improves degree of asymmetry in infants with postural asymmetry

<https://link.springer.com/article/10.1007/s00381-016-3200-5>

EFFECTS OF OSTEOPATHIC MANIPULATIVE TREATMENT ON CHILDREN WITH
PLAGIOCEPHALY IN THE CONTEXT OF CURRENT PEDIATRIC PRACTICE: A
RETROSPECTIVE CHART REVIEW STUDY
JOM, DECEMBER 2023

Results

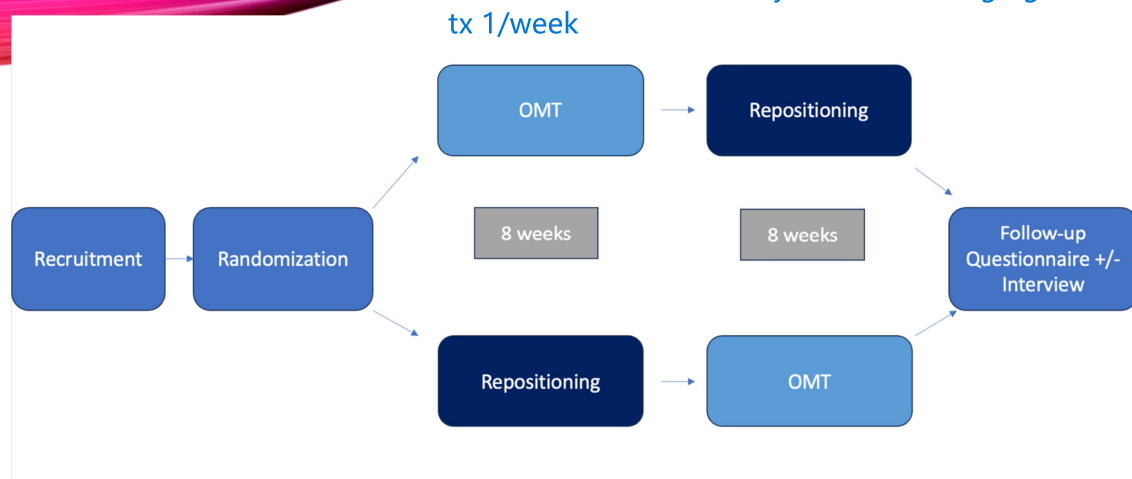
Participants demonstrated a mean CVAI – a measure that determines the severity of DP – of 6.809 (± 3.335) (Grade 3 severity) at baseline, in contrast to 3.834 (± 2.842) (Grade 2 severity) after a series of OMT treatments. CVAI assessment after OMT reveals statistically significant ($p \leq 0.001$) decreases in measurements of skull asymmetry and occipital flattening. No adverse events were reported throughout the study period.



**OPT-IN:
OSTEOPATHIC
PLAGIOCEPHALY
TREATMENT FOR
INFANTS AND
NEONATES**



tummy time, encouraging looking opposite side



Outcomes of Interest

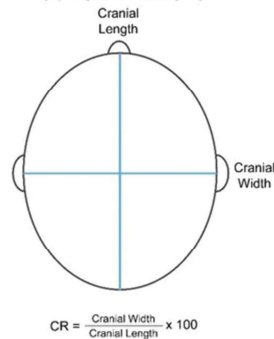
Primary Outcomes:

- Cranial Vault Asymmetry Index (CVAI)
- Head circumference
- Cephalic Ratio

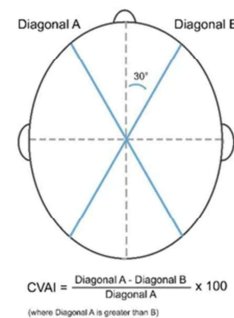
Secondary Outcomes (evaluated at ~ 1-year):

- Developmental milestones/history
- Medical history - cases of otitis media, etc.
- Developmental delay
- Allergic rhinitis
- Vision problems (e.g., strabismus)
- Torticollis
- Hospitalizations
- Use of helmet after study concluded
- Colic
- Reflux/GERD

(A) Cephalic Ratio (CR)



(B) Cranial Vault Asymmetry Index (CVAI)



not just a cosmetic issue is functional

SNEAK PEEK



	Cranial Index (CI)	Cranial Vault Asymetry Index (CVAI)
06-13-2024	80.7 ± 0.2	+5.9 ± 0.4 Positive CVAI means the right diagonal was shorter (right flattening)



	Cranial Index (CI)	Cranial Vault Asymetry Index (CVAI)
06-13-2024	78.3 ± 0.3	+2.8 ± 0.5 Positive CVAI means the right diagonal was shorter (right flattening)

mild-mod plagiocephaly

immediate pre and post omt measurements



OBJECTIVES

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:
 1. Plagiocephaly/torticollis and feeding dysfunction
 - 2. Otitis Media**
 3. Concussion
2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study
3. Compare and contrast normal motion and pathological motion of the individual cranial structures potentially associated with conditions discussed during the self-study.
4. Define relevant research as it relates to the conditions emphasized in the self-study.



OTITIS MEDIA



more lymphatic focus

CASE 2

- 18-month-old F presents for a sick visit with dad. Dad reports that the child has been pulling at her ear – AGAIN! Started tugging at ears again yesterday but had several ear infections over the last year. She has also had a fever of 101° F since last night. They have been giving ibuprofen for the fevers and discomfort with some relief. No ear drainage. Dad thinks this is the child's fifth ear infections total, and the most recent was one month ago. The episode last month was treated with a 10-day course of amoxicillin. Dad thinks the longest the child has gone without an ear infection has only been a couple months. He also reports that the child always seems to have a runny and stuffy nose. Has not been teething recently, and already has his 12-month molars. Have only seen their PCP about these infections. Dad is also worried that she does not hear him all the time, and feels he has to turn the TV up louder than he used to when she is watching.

CASE 2 CONTINUED

- Birth history: born at 41 weeks via C-section for failure to progress during labor. Pregnancy was otherwise uncomplicated, and baby did well after delivery.
- Feeding history: Eats a regular (picky) toddler diet. Was breastfed for first 1-2 months, but then mom went back to work and they transitioned to formula. Stopped formula at 12 months and transitioned to whole milk.
- ROS: + ear pain, fever, congestion, rhinorrhea, fussiness, ear pulling. Negative for eye redness or drainage, otorrhea, feething.
- Meds: acetaminophen and ibuprofen prn
- Vaccinations: Up to date by report
- Allergies: NKDA
- PMHx/PSHx: no significant PMH other than ear infections as per HPI, no surgeries
- Trauma: none reported other than birth
- Family Hx: Mother and Father both alive and well and both have allergic rhinitis. Has one healthy brother.
- Social Hx: Lives with mom, dad, and brother. Goes to daycare 3 days a week. Both parents smoke outdoors. One dog in the home, but it is not new.
- Developmental history: only saying four words – mama, dada, hi, bye
normal should be saying 6-10 words

PE– PERTINENT POSITIVES

- Vitals: 99.6° F HR 120 RR 25 BP 90/60 BMI: 50%
- Growth charts appropriate for height, weight, and head circumference
- Gen: Alert, WDN, fussy but consoles easily in dad's arms
- Head: NCAT, AF closed, no plagiocephaly assoc with allergic rhinitis
- Eyes: no conjunctival injection, no discharge, bilateral allergic shiners
- Ears: bilateral TM with effusions – purulent on the left, serous on the right; bulging TM on the left with erythema
- Nose: pale, boggy turbinates, clear rhinorrhea
- Mouth/throat: moist mucus membranes, no tonsillar enlargement or exudate, cobblestoning in posterior oropharynx

know when last dose of ibuprofen was given

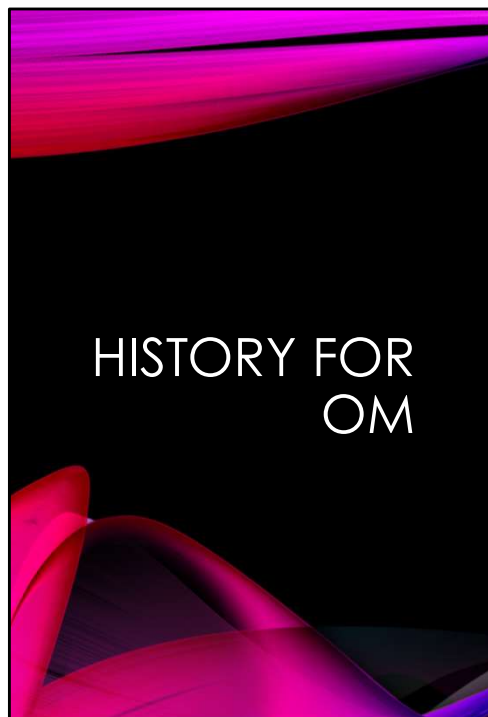
OSTEOPATHIC STRUCTURAL EXAM

- Head: OA restriction ($FR_R S_L$), diminished CRI, boggiess over bilateral maxillae without tenderness, internally rotated temporal bone on the left, compressed OM suture L>R
- Cervical spine: hypertonicity in paraspinal musculature, congestion over bilateral SCM but worse on the left, $C2FR_L S_L$
- Thoracic spine: thoracic inlet restricted-rotated right and sidebent right, T1 $FR_R S_R$, boggiess and hypertonicity T1-4 on the right
- Ribs: elevated first rib on the left
- Abdomen: thoracoabdominal diaphragm restriction



WHAT ARE YOUR DIAGNOSES?

- Allergic rhinitis
- Acute and recurrent otitis media on the left
- Serous otitis media on the right
- Concern for hearing loss
- Speech delay



HISTORY FOR OM

Risk factors: allergies, craniofacial abnormalities,
family history, formula feeding, SHS exposure

How many episodes of OM? [need other intervention?](#)

How have they been treated?

When was the last one?

Seen ENT?

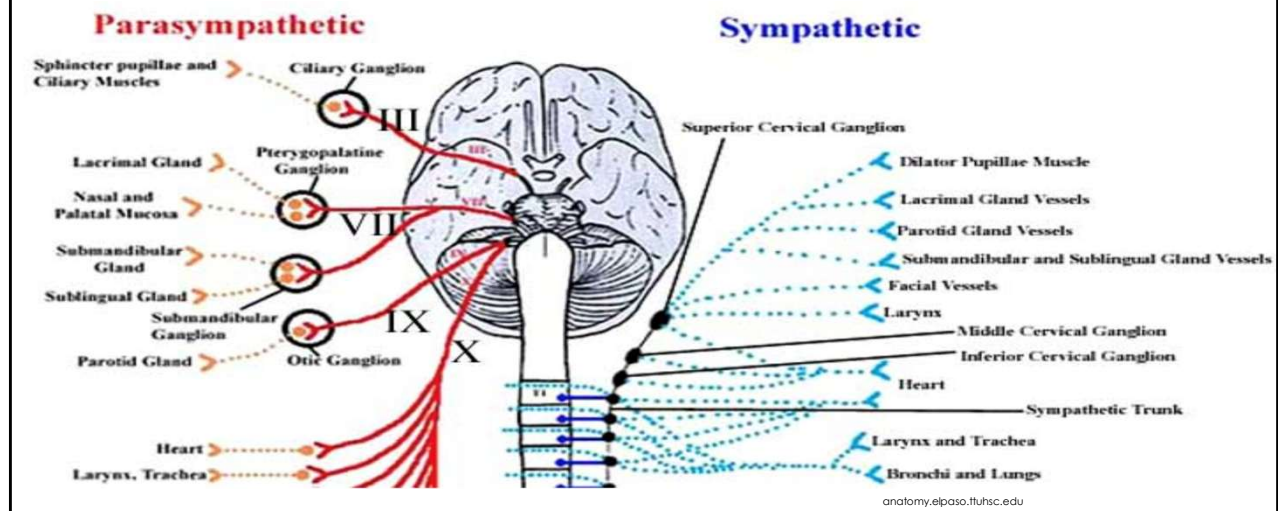
Otorrhea?

Hearing changes?

Speech delays?

[immune advantages with breastfeeding](#)
[sucking when breast feeding has diff muscle activation](#)

ANATOMIC CONSIDERATIONS – AUTONOMICS



Pterygopalatine

T1-T4 S-ANS to head

CN 3, 7, 9, 10 P-ANS to head

PTERYGOPALATINE GANGLION

Parasympathetic

Innervated by petrosal
(branch of CN7)

Axons go to lacrimal glands,
soft palate, tonsils, uvula,
upper lips, gums, upper
pharynx, and nasal mucosa

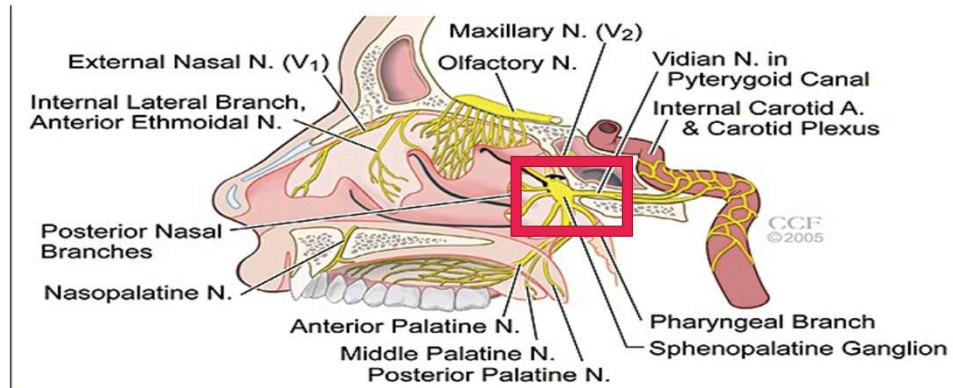
Are some post ganglionic
sympathetic fibers in the
ganglion as well

If treated can lead to
vasodilation and increased
drainage, thinner secretions

Control blood flow to the
conchae

edema in nose

PTERYGOPALATINE GANGLION



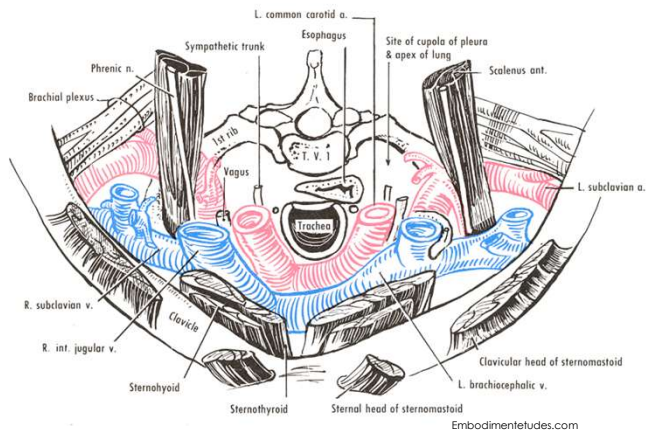
can tx with omt

ANATOMIC CONSIDERATIONS - BIOMECHANIC

Thoracic inlet
restriction

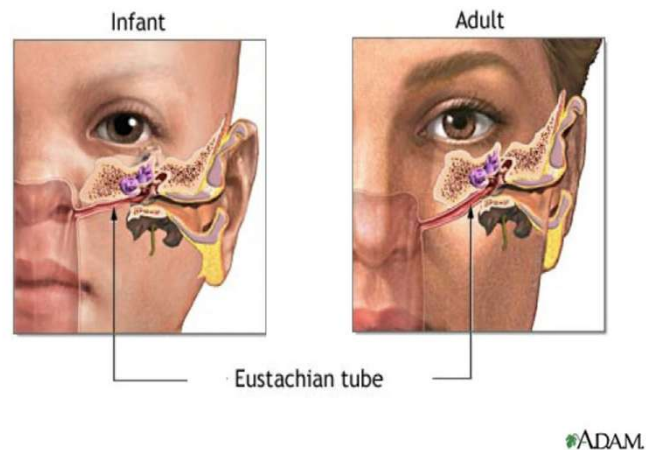
Cervical
congestion

- Especially over anterior
cervical chain **along scm**



EUSTACHIAN TUBE

- Kids vs adults
 - More pliable and easily compressible
 - More horizontally oriented
 - Shorter
 - Poor ability to equilibrate pressure
- Changes after ~7 y/o
- Proximal 1/3 is bony, remainder is fibrocartilaginous
- Muscles: tensor veli palatini, levator veli palatini, salpingopharyngeus, tensor tympani



Adult angle is 45 degrees and kids is 10 degrees

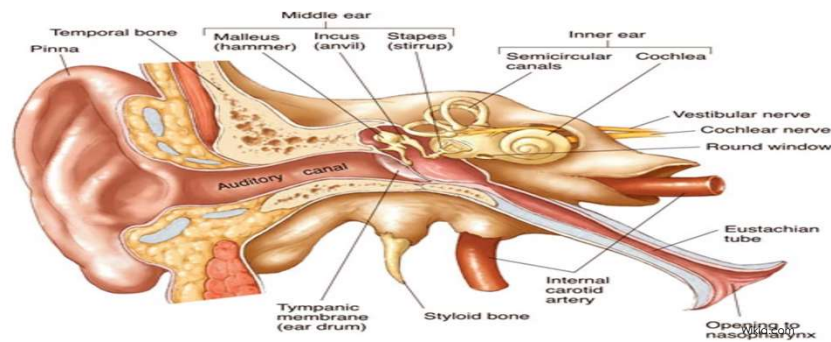
Also feed supine, reflux, etc

10 mm infants, 36 mm adults

easier to close and prevent drainage - risk for recurrent infections with stasis of fluid, hearing issues - > speech delay

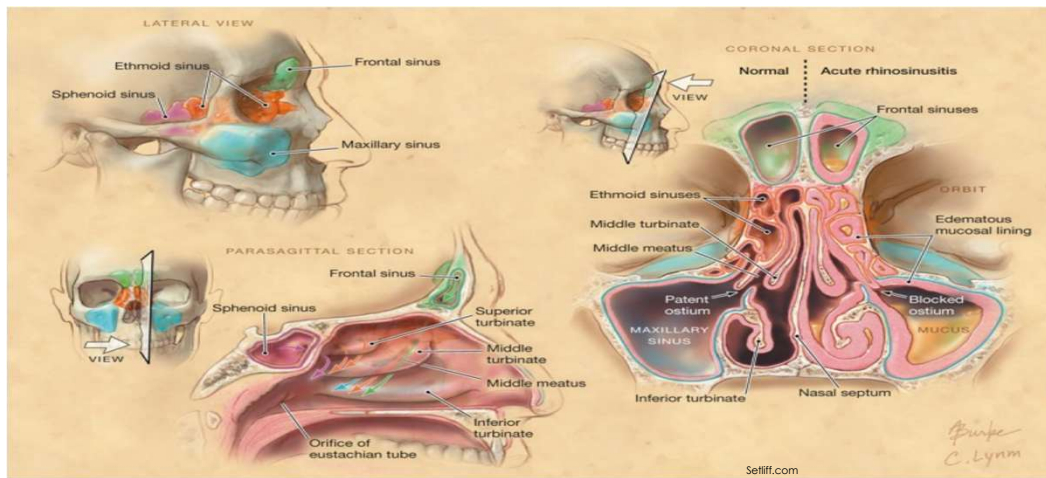
OSSEUS CONSIDERATIONS

- The middle/inner ear and bony portion Eustachian tubes lie within temporal bone
 - Squamous and petrous portions
- Distal Eustachian tubes pass through groove between petrous portion temporal bone and greater wing of sphenoid

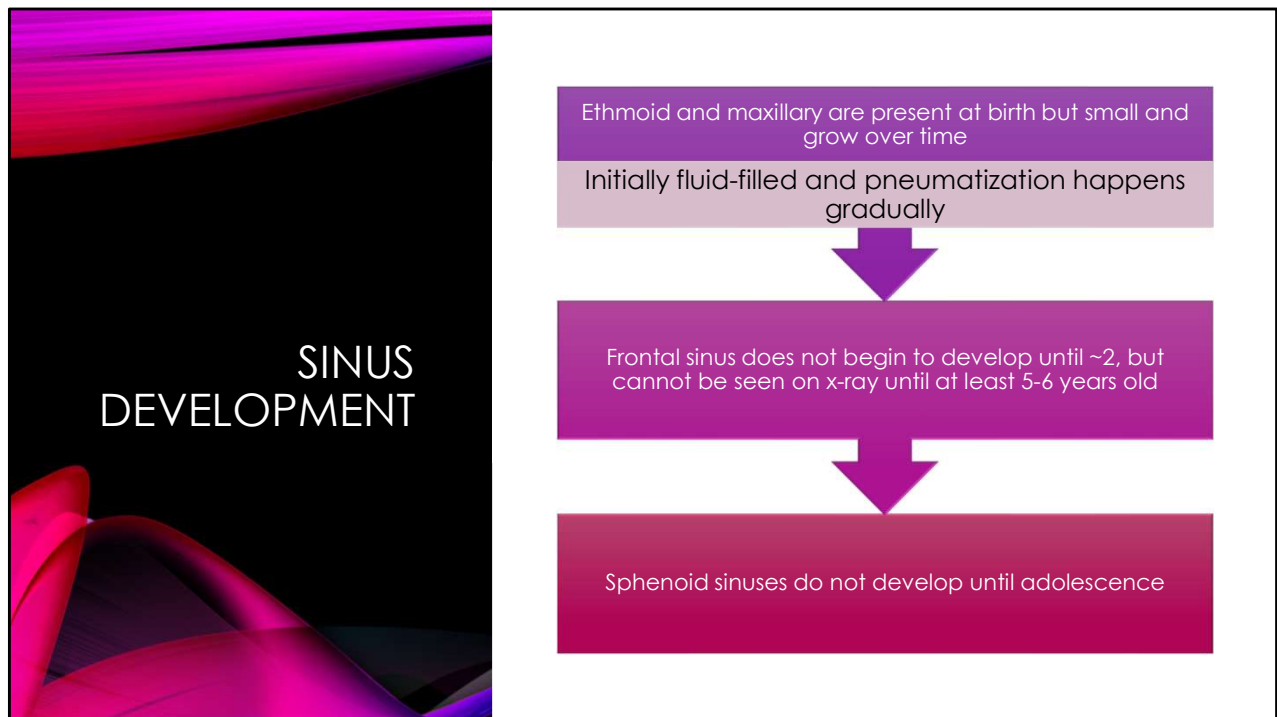


ANATOMIC CONSIDERATIONS – OSSEUS

- Sinuses in maxillary, ethmoid, sphenoid, and frontal bones



ethmoid and maxillary present at birth



diff anatomy depending on a



CIRCULATION/LYMPHATICS

- All about sequence!
 - Start with thoracic inlet
 - Left before right...why?
 - Then move proximal for what you would like to drain next
 - If we ultimately want to drain heads – where do you start?

OSTEOPATHIC RESEARCH

- **J Aller Ther- An Osteopathic approach to chronic sinusitis**
 - Pilot study with 16 patients
 - Reported improved pain with OMT
- **JOM – Osteopathic evaluation and manipulative treatment in reducing the morbidity of otitis media: a pilot study**
 - Only 8 patients enrolled, but most of the patients had no recurrence of symptoms
- **JOM – Effect of Osteopathic Manipulative Treatment on Middle Ear Effusion Following Acute Otitis Media in Young Children: A Pilot Study**
 - 43 patients enrolled
 - Reduced duration of effusion following AOM with OMT

most sig study

small pts and method flaws but overall positive results



Gong to cover extensive anatomy and is meant to be more of an overview – don't get bogged down by details but realize the far reaching effects that OCMM can have on these patients. Concussions have complicated and varied presentations – multiple SXS.

You have covered a lot of this anatomy previously including OCMM 4 and 5.



OBJECTIVES

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:
 1. Plagiocephaly/torticollis and feeding dysfunction
 2. Otitis Media
- 3. Concussion**
2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study
3. Compare and contrast normal motion and pathological motion of the individual cranial structures potentially associated with conditions discussed during the self-study.
4. Define relevant research as it relates to the conditions emphasized in the self-study.

CASE 3

- 16-year-old cisgender male presenting with headaches that started after a concussion 2 weeks ago. The headaches are constant, but do wax and wane throughout the day. They are localized to "the whole head." HA worsened by light, schoolwork, using electronics, sounds, and activity. Rest in a dark room helps the most to relieve the HA. Ibuprofen 800 mg has also given some relief, but upsets the patient's stomach so she does not take it often.
- He rates the HA currently a 5/10 which is average, but they can worsen to an 8/10 at times. Pt also has photophobia, phonophobia, decreased appetite, feels more tired yet finds it difficult to fall asleep, and per mom has also been "grumpier than normal." Pt also reports some dizziness and blurry vision bilaterally which have both been intermittent since the concussion. He also reports some neck pain that started after the injury and occasionally hears ringing in his ears.
- Trauma history: Pt suffered their first concussion 2 weeks ago while playing in a football game. Had a helmet to helmet hit. Pt does not really remember the incident but thinks that the other player hit the front of his helmet. He did briefly lose consciousness and was "out of it" for a few minutes when he woke up. His parents took him immediately to the local ER. They examined him and diagnosed him with a concussion. He was not imaged. He was discharged home with supportive care instructions including restricted activities and no football allowed.
 - Also had a broken right femur when he was 12 while playing football

CASE 3 CONTINUED

- Birth history: full-term, SVD, uncomplicated per mom
- Diet: Regular diet
- ROS: + headaches, anorexia, photophobia, phonophobia, vision changes, sleep disturbances, tinnitus, mood changes, fatigue
- Meds: 800 mg ibuprofen prn; cetirizine prn – usually in springtime
- Vaccinations: Up to date by report
- Allergies: NKDA
- PMHx/PSHx: history of allergic rhinitis; had ORIF of his right femur 4 years ago
- Family Hx: Mother and father both alive and well and have no significant history. One sister with migraines.
- Social Hx: Lives with mom, dad, and sister. Doing well in HS, but grades suffering the last couple weeks because its hard to concentrate. Wants to be an engineer when he grows up. Denies tobacco, EtOH, marijuana, or illicit drug use. Sexually active with one female partner and uses condoms.
- Developmental history: no history of developmental delays.

CASE 3

- VSS (vital signs stable)
- HEENT: EOMI, PERRLA, normal fundoscopic exam, Snellen 20/20 bilaterally
- CV/Lungs: Normal
- Neuro: CN II-XII grossly intact, DTR 2/4 b/l in UE and LE, MS 5/5 in UE and LE b/l, No atrophy, spasticity, clonus or tremors noted. Normal finger-to-nose, Romberg and heel-to-shin. Sensation and two-point discrimination intact. Proprioception intact. (normal!)
- OSE:
 - Head: SBS compression with no appreciable PRM motion, bilateral frontals externally rotated, OA FS_LR_R hit near metopic suture likely
 - Cervical: Tenderness/hypertonicity in suboccipital region and bilateral paraspinal cervical musculature, C3 FR_RS_R
 - Thoracic: Paraspinal muscle hypertonicity and tenderness in upper thoracic region, hypertonic trapezius worse on the right
 - Lumbar: Paraspinal muscle hypertonicity
 - Sacrum: diminished motion (PRM)



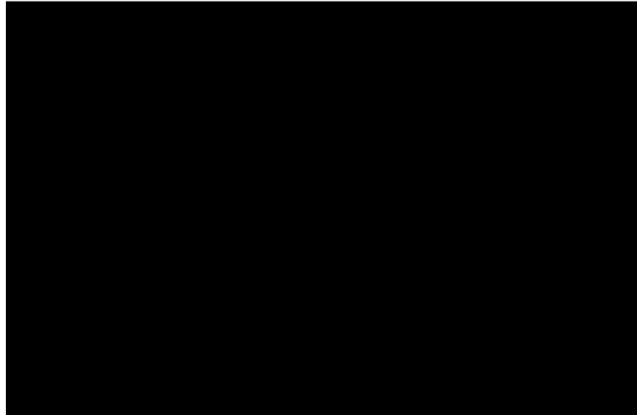
PROBLEM LIST

- Concussion
 - Headaches
 - Anorexia
 - Mood changes
 - Sleep disturbances
 - Tinnitus
 - Vision changes
 - Photophobia
 - Phonophobia
 - Neck pain
 - Fatigue
 - Difficulty concentrating (grades dropping)



WHAT HAPPENS TO THE BRAIN WITH A CONCUSSION

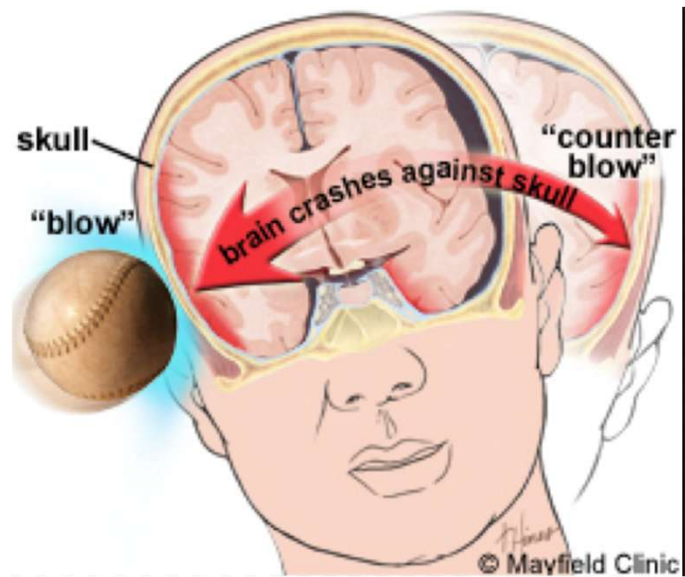
[Your Brain on Concussion](#)



<https://www.nytimes.com/interactive/2017/01/09/sports/football/what-happened-within-this-players-skull-football-concussions.html?ribbon-ad-idx=5&src=trending&module=Ribbon&version=context®ion=Header&action=click&contentCollection=Trending&pgtype&mtrref=undefined&gwh=58D5A76521D97D95373F1038596274BB&gwt=pay&assetType=PAYWALL>

ETIOLOGY OF SYMPTOMS

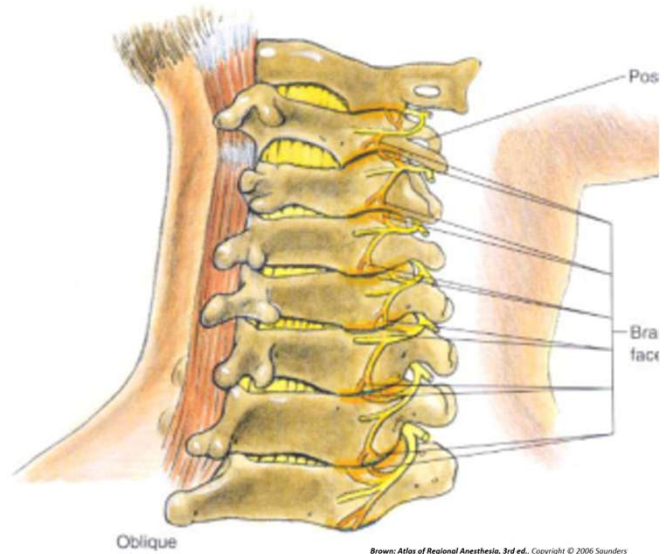
- Complex mechanisms
 - Pain from direct primary trauma
 - Inflammation and edema cause pressure on dura
 - Trigger intracranial mechanical pain generators
 - Secondary (indirect) trauma from inertia
 - Cervical spine
 - Sacrum

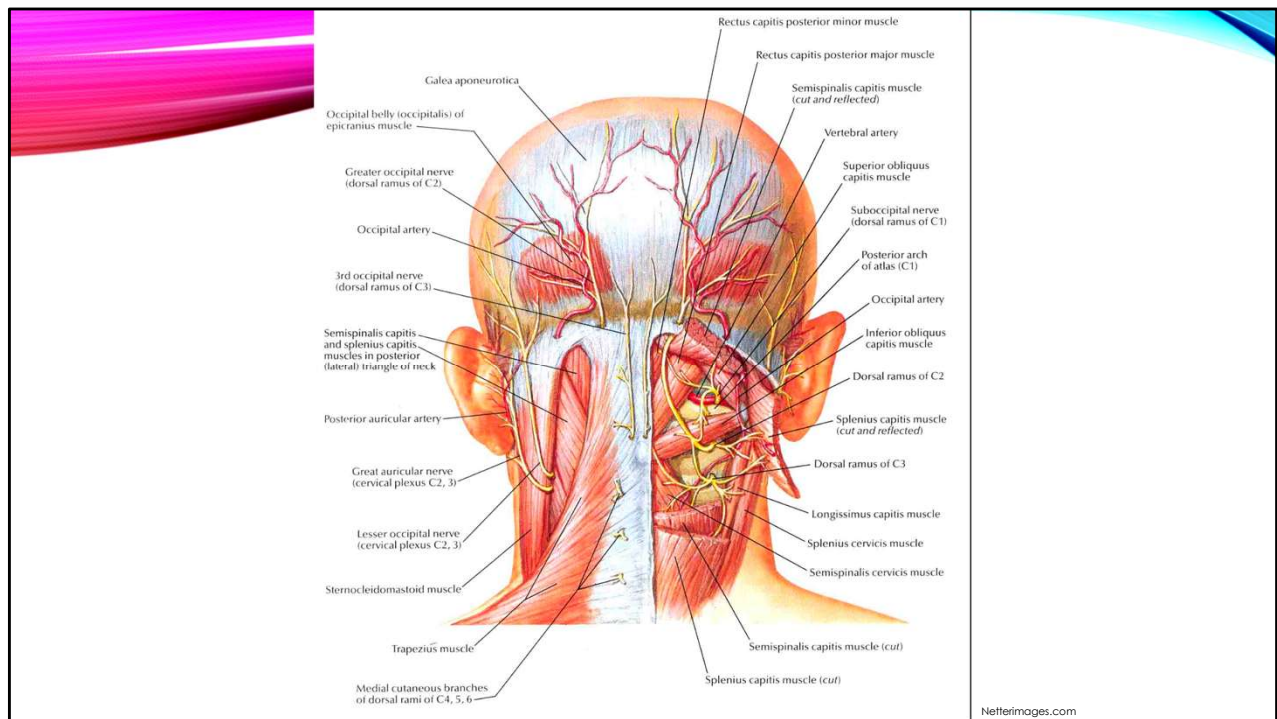


focus on what is pertinent for ocmm - where you might localize for specific bones

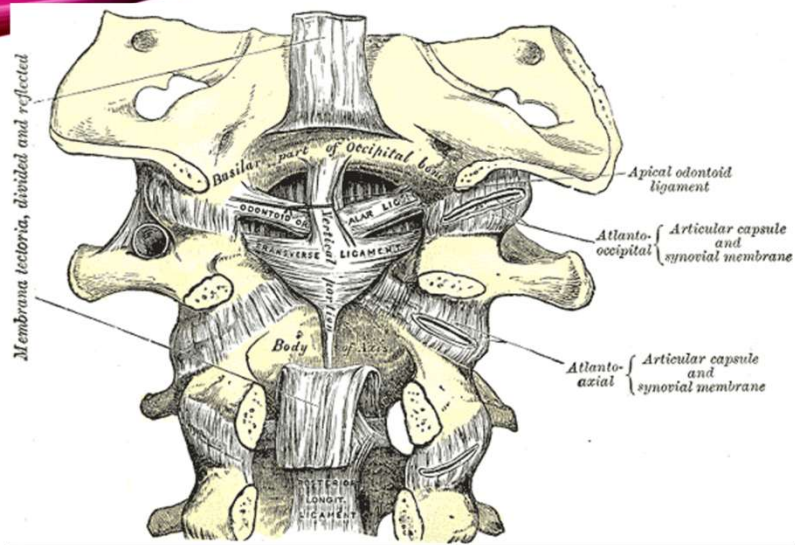
CERVICAL SPINE

- Pain generators
 - Myofascial tissue
 - Nerve compression or discogenic impingement
 - Ligamentous strains
 - Facet joints –innervated by medial branch of dorsal ramus of spinal nerve
 - Each joint supplied by its own level, and the one above
 - Pain referred to head, cervical spine, thoracic spine – reproduced from extension and rotation





radiating up to head from cervical spine - why can lead to headache and head pain



Boneandspine.com

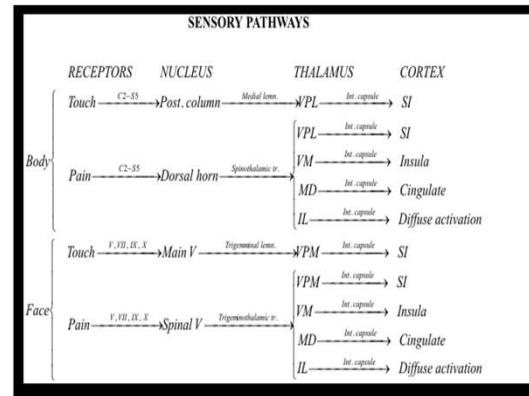
CNS FACILITATION

Nociception from soft tissues in upper cervical spine carried through the anterolateral system or spinothalamic tract

- Anterior portion carries information about crude touch/pressure
- Lateral portion carries information about pain and temperature

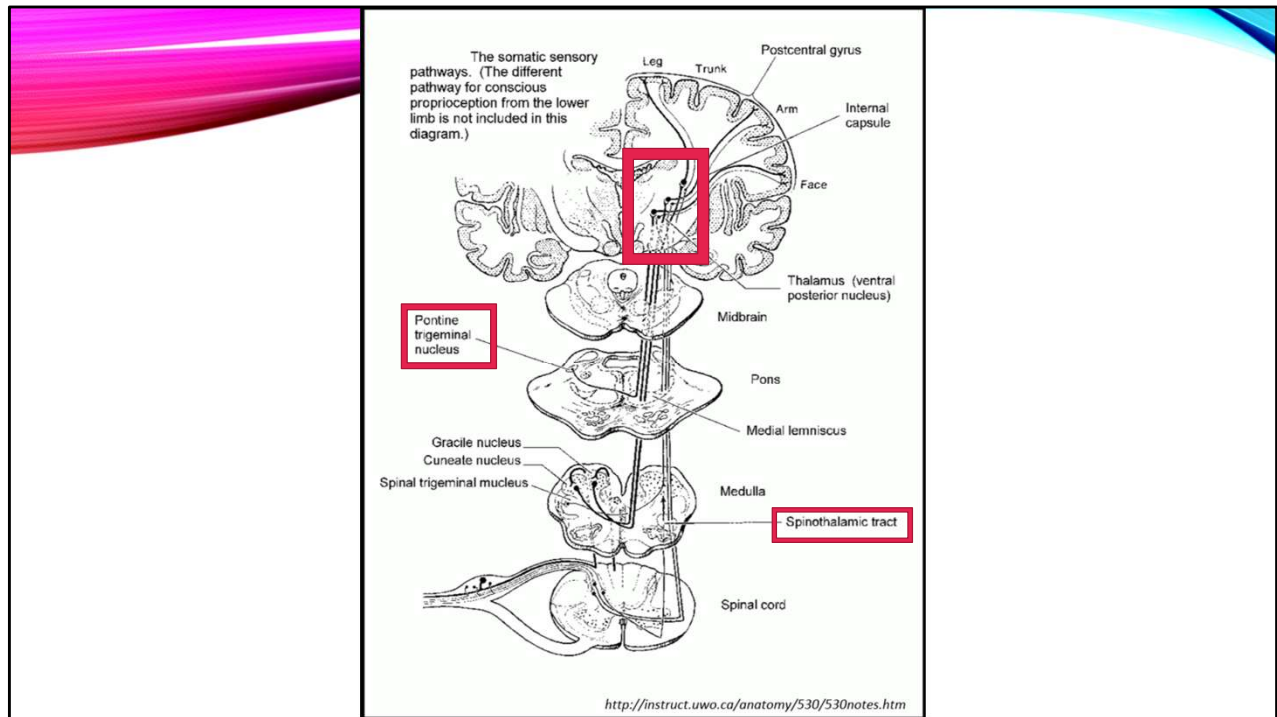
ALS and trigeminal overlap

- Cervical strains → headaches and associated symptoms



Wikipedia.org

cervical can lead to headaches where CN V is originating



DIZZINESS/BALANCE ISSUES

Cervical spine dysfunction → altered input to cerebellum

- Postural reflexes (cervico-ocular and vestibulo-ocular reflexes) play a role as well → altered motor output

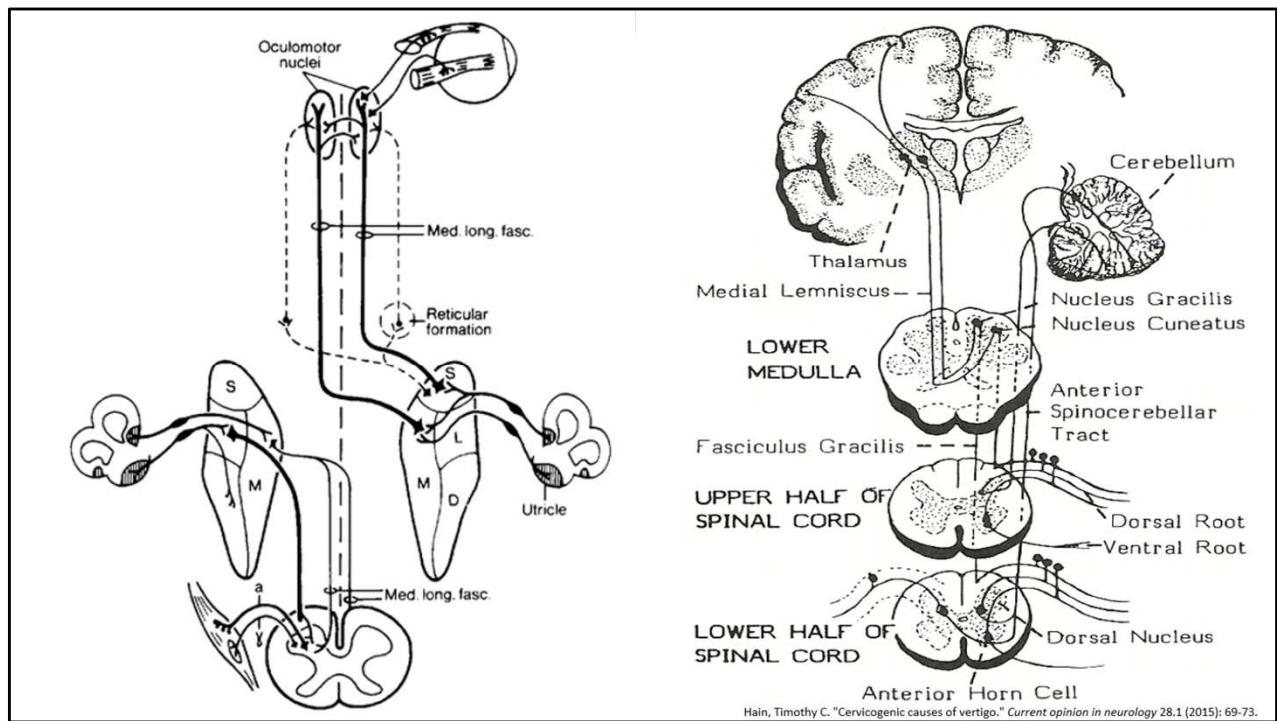
Cranial base dysfunction → Temporal bone motion altered

- Otoliths disturbed (tinnitus)
- Postural reflexes
 - Vestibulospinal reflex, vestibulo-ocular reflex, vestibulocolic reflex

petrous and mastoid portion of temporal

Vestibular changes

- Acceleration/deceleration from trauma may cause mild prolonged “motion sickness” due to changes in sensory cells of membranous labyrinth



HEARING AND VISION PATHWAYS

Upper cervical dysfunction →
facilitation → SNS activates
superior cervical ganglion →

- Iris → Altered accommodation
- Middle ear → ↓ Blood flow → effusions, tinnitus, hearing loss

vertebral to cochlear artery

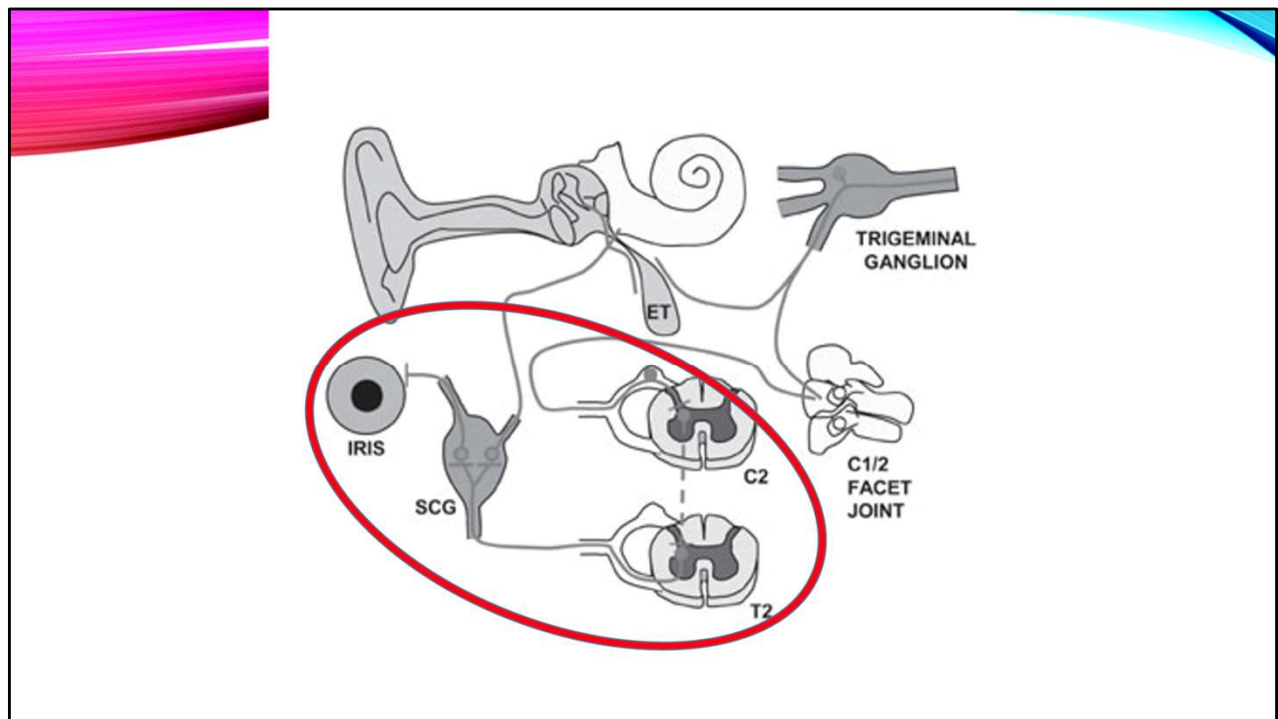
Upper cervical dysfunction →
facilitation of CN5 →

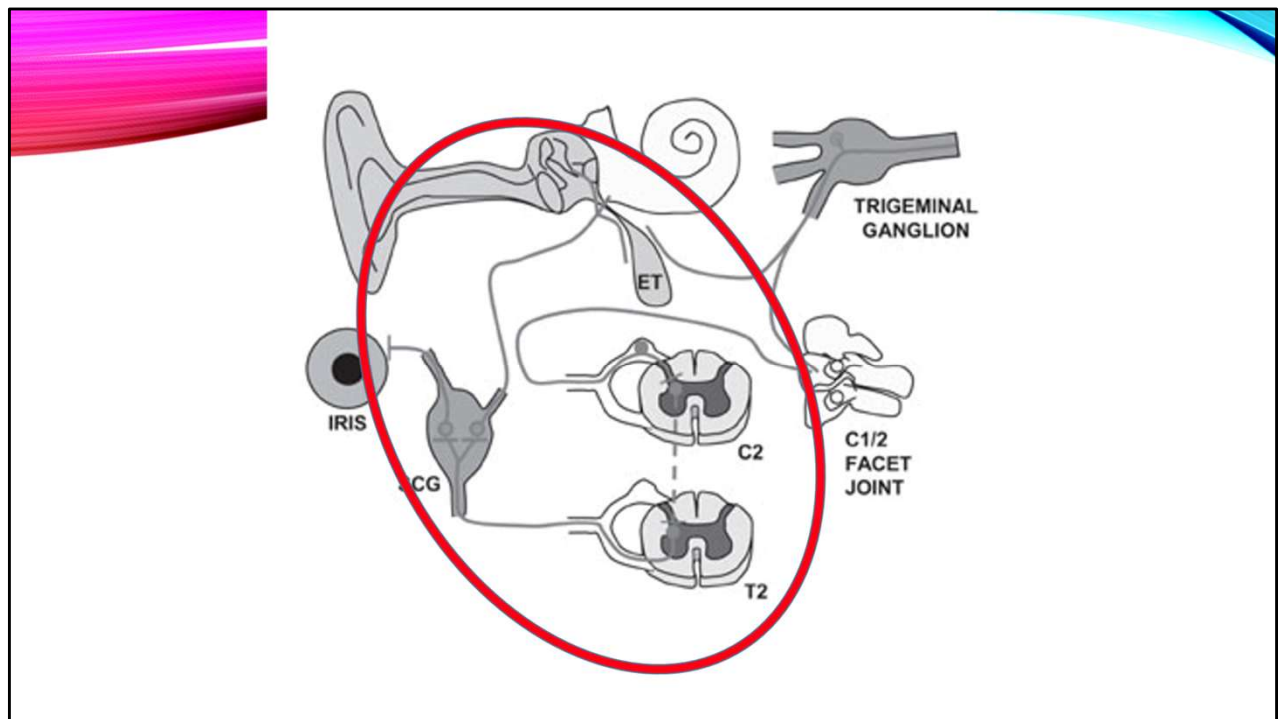
- Tensor veli palatini → Eustachian Tube Dysfunction
- Tensor tympani inhibiting accommodation to loud noise → phonophobia

can't modulate sound

altered sound
processing and
hearing

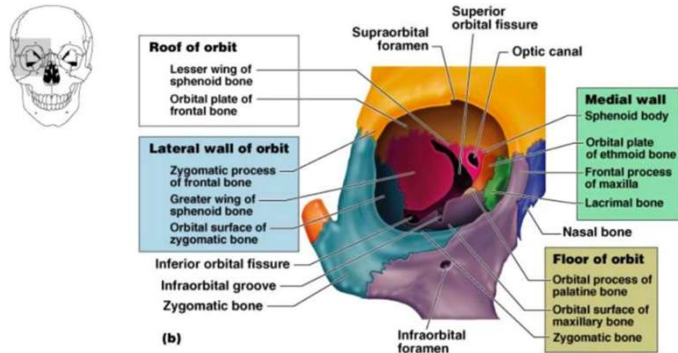
Decreased blood flow is from vertebral to cochlear artery and arterial constriction



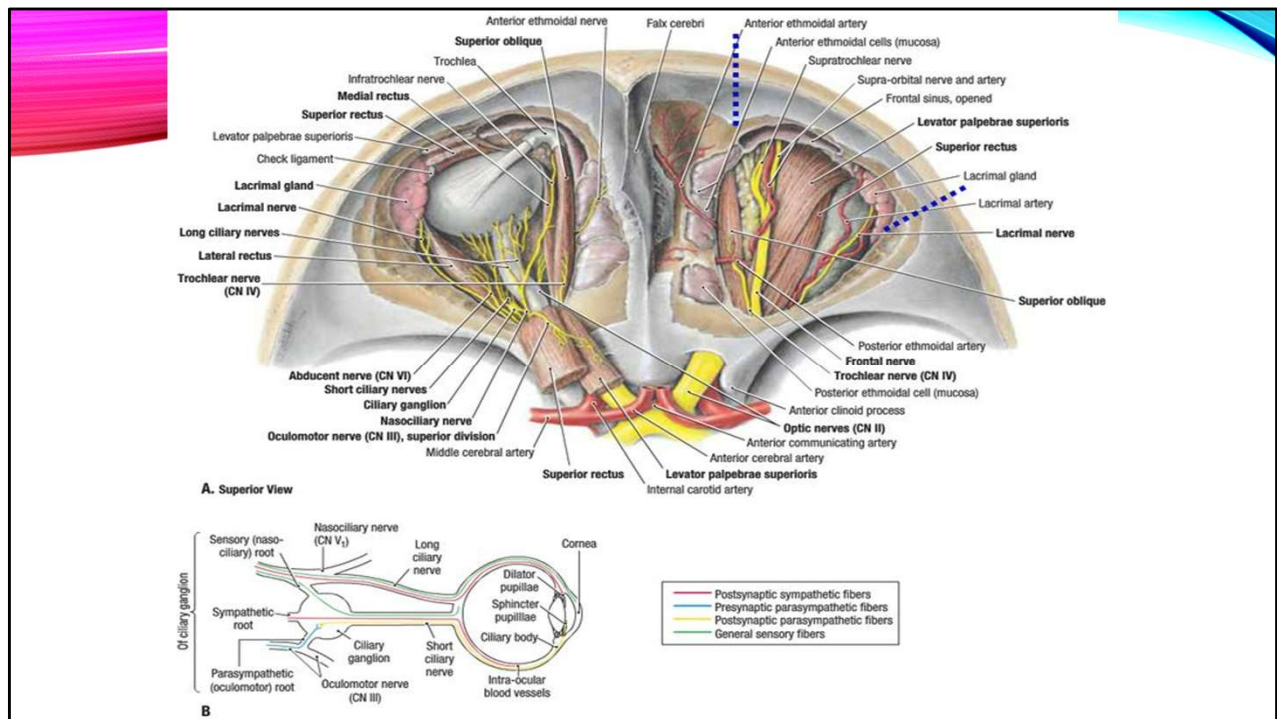


THE ORBIT

- Frontal bone
- Sphenoid
- Ethmoid
- Lacrimal bone
- Palatine *often forgotten*
- Maxilla
- Zygoma



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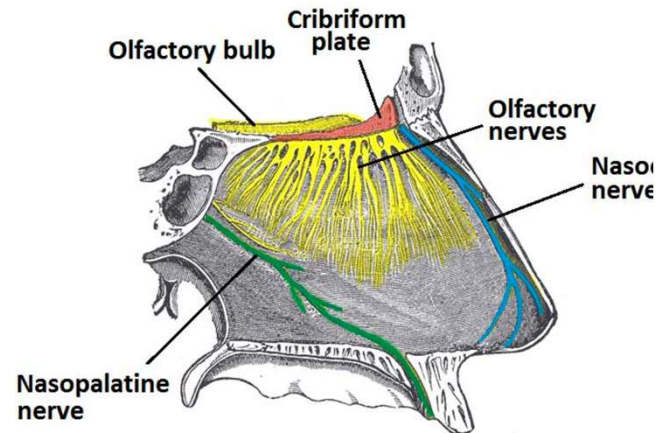


CN 2,3,4,6

SMELL ALTERATION

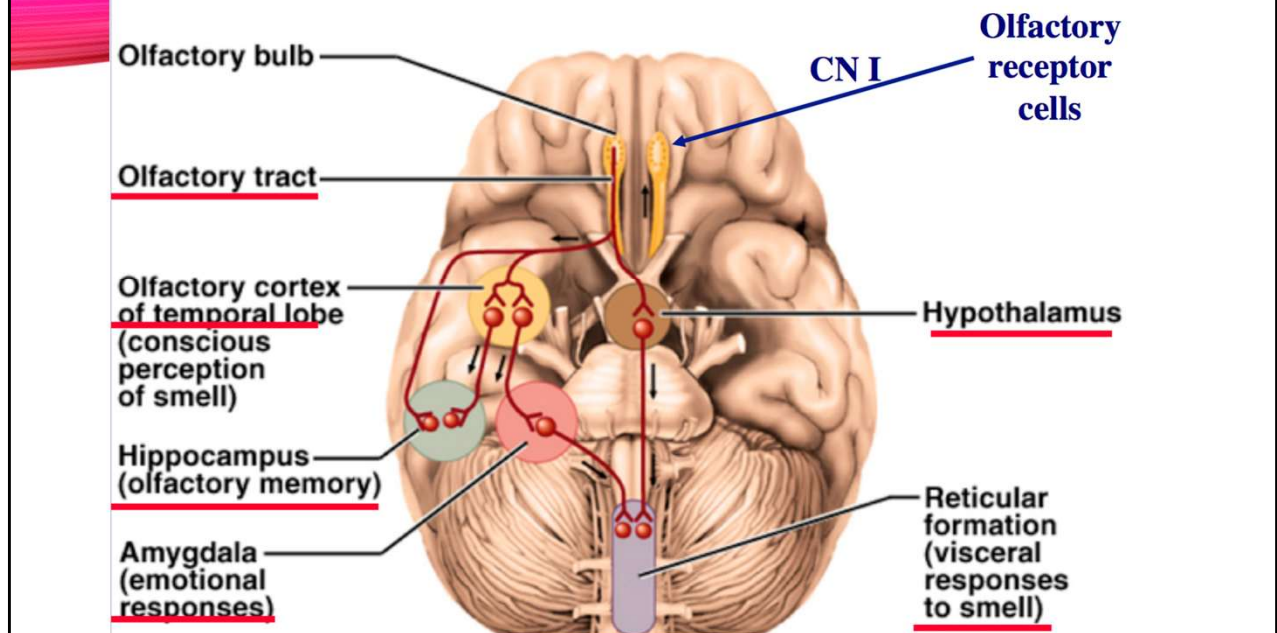
- Anosmia – partial or complete
 - Affects taste
 - Can be due fracture of cribriform plate (ethmoid)
 - Occipital or parietal trauma can also damage olfactory bulb tracts

can lead to anorexia



© teachmean
The Art of Learning

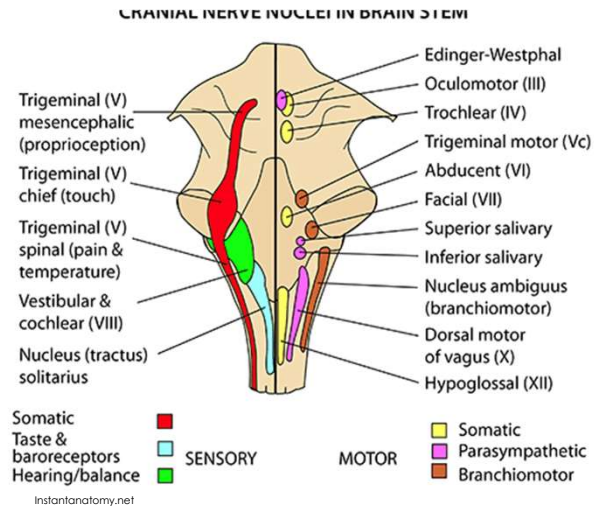
Olfactory Projection Pathways

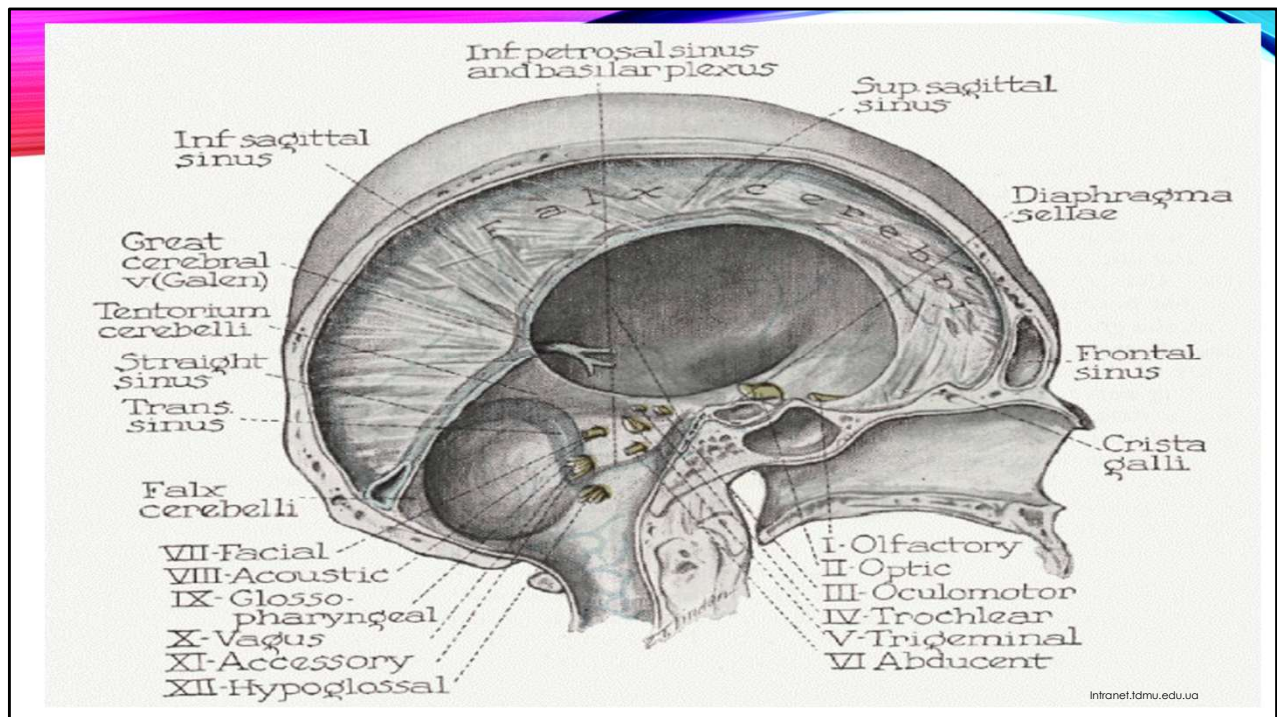


FACILITATION OF CN

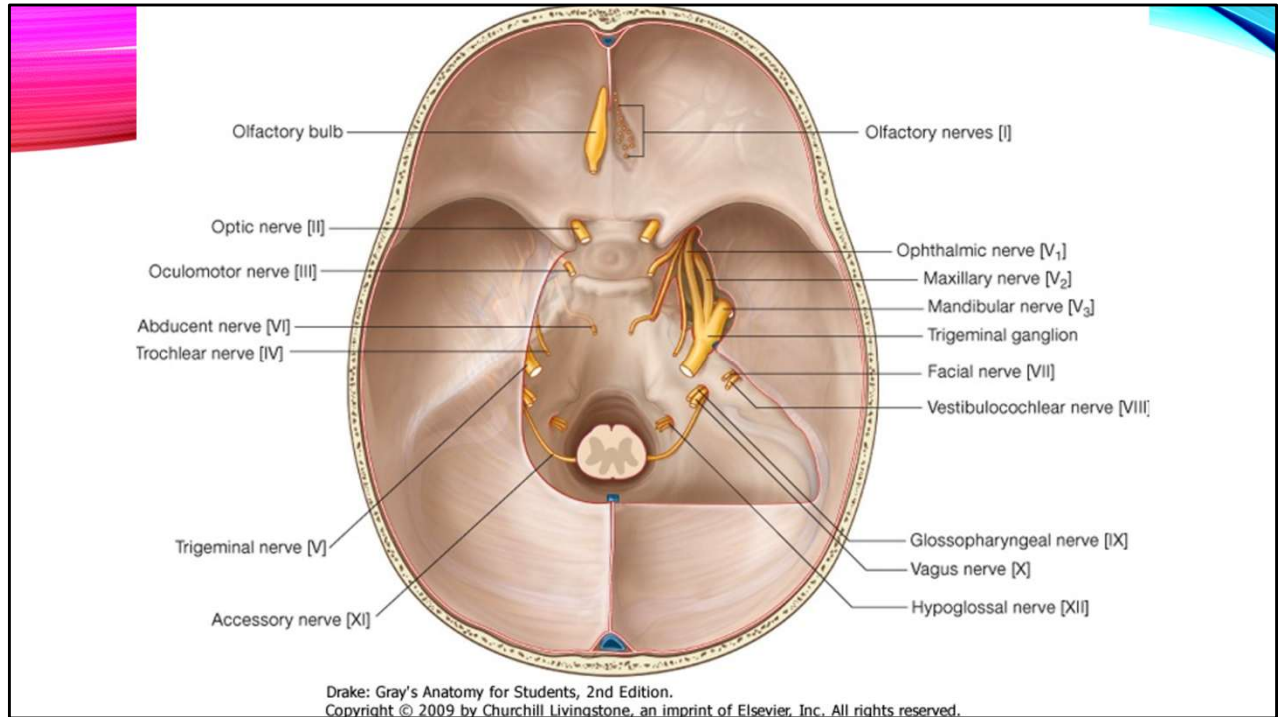
- Facial Nerve: facial pain
- Glossopharyngeal and vagus: nausea
- Vagus: dysphagia and GI disturbance, appetite changes
- Spinal accessory: Neck stiffness and pain

SCM and trap



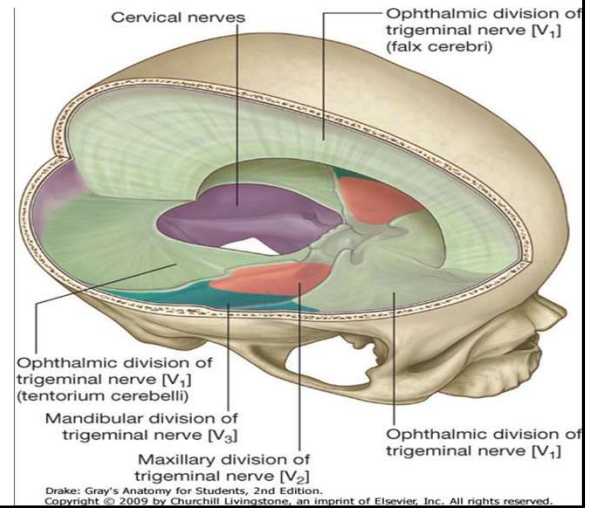


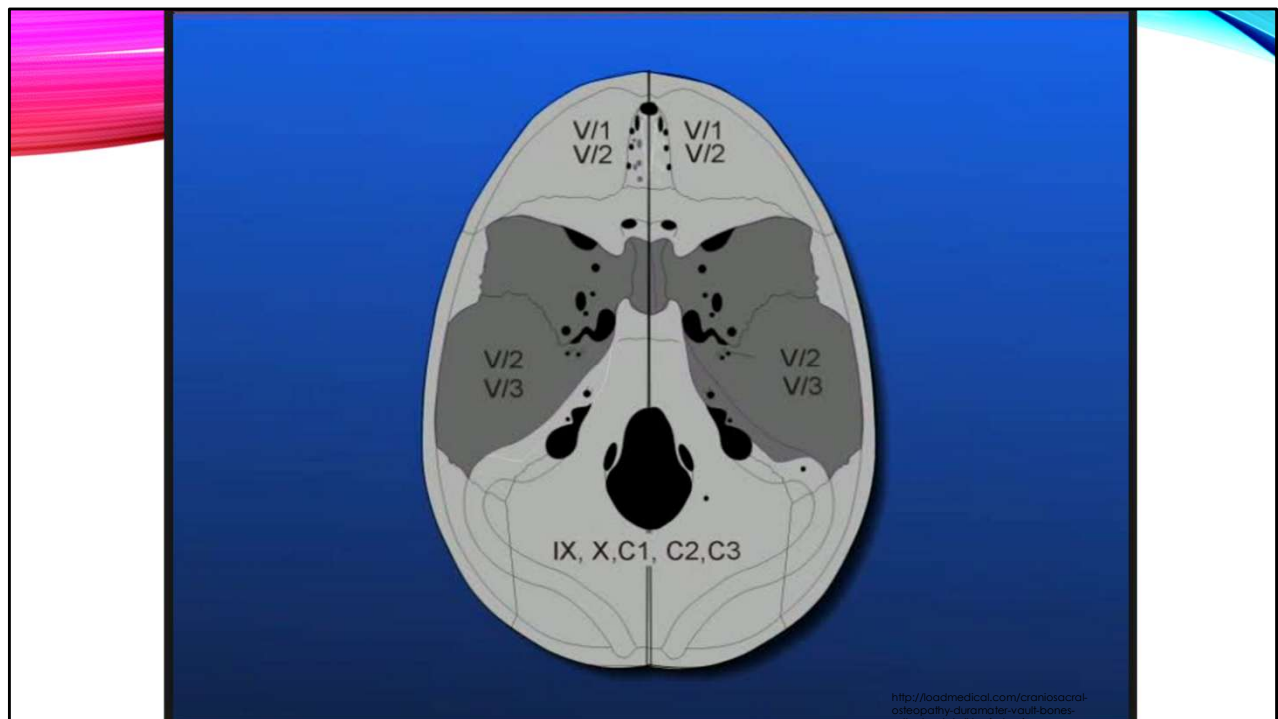
dura not allowing flexibility - influencing function of nerves



INNERVATION OF DURA

- No true pain fibers in brain parenchyma
- Meningeal pain fibers
 - Anterior and middle cranial fossa from CN5
 - Posterior cranial fossa: C1-3 via CN9 & 10



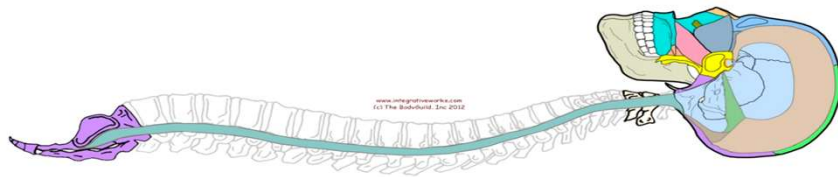


SO WHAT?

- These intimate relationships explain how even slight disruptions in the system can result in the wide variety of symptoms
 - Nausea, vomiting – CN X
 - Dizziness – CN VIII, X
 - Tinnitus – CN VIII
 - Decreased appetite – CN X

OTHER THOUGHTS ON DURA

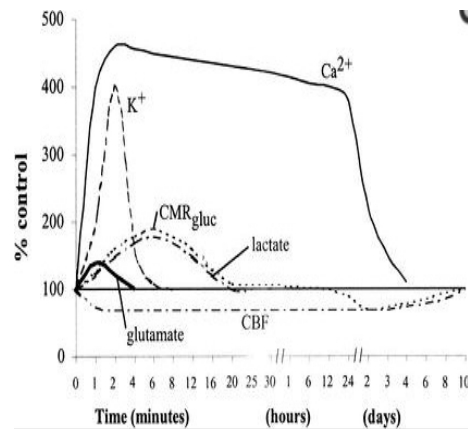
- These attachments outside of the head, help explain why the trauma to the head may lead to pain elsewhere in the body
 - Neck pain
 - Back pain
 - Sacral/pelvic pain
 - And ultimately further impaired functioning



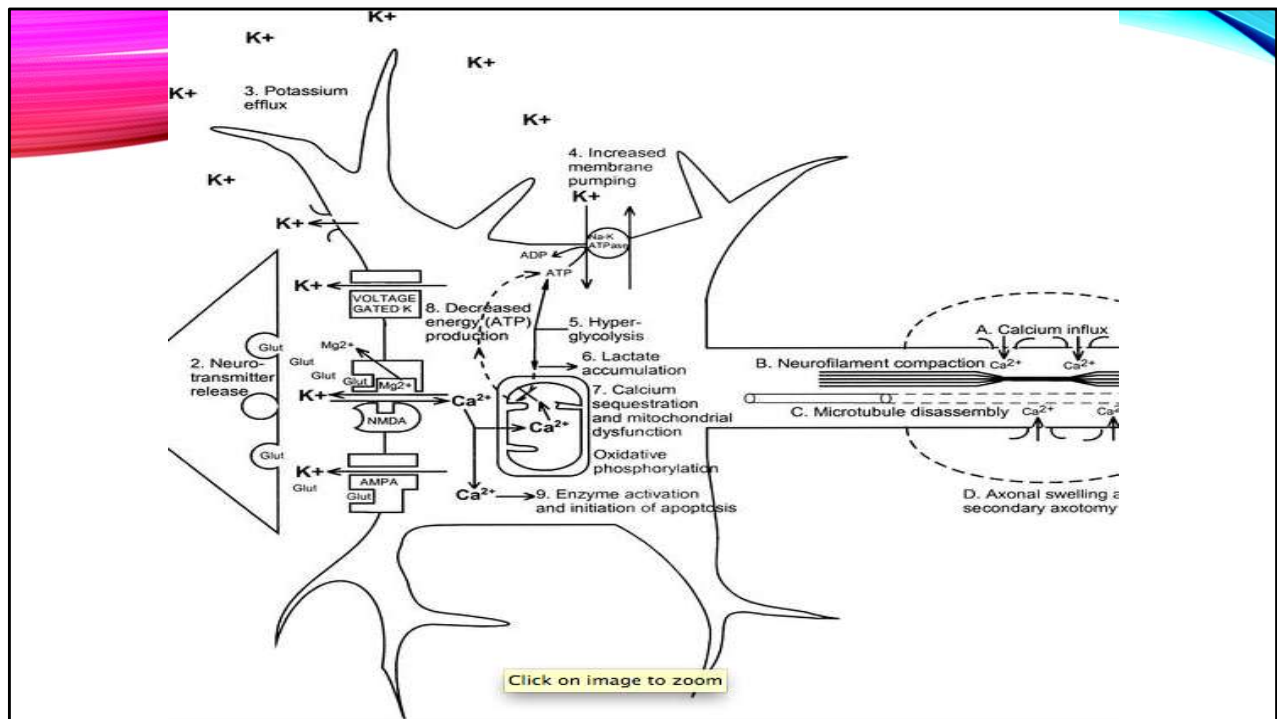
foramen magnum is C2-C3 then to sacrum

ANYTHING ELSE?

- Studies have shown there is a pathophysiologic cascade following concussion
 - Neuronal depolarization
 - Release of excitatory neurotransmitters (glutamate)
 - Ionic shifts (K, Ca, Na, Mg)
 - Changes in glucose metabolism (to meet demands of shifts)
 - Altered cerebral blood flow (energy crisis!)
 - Impaired axonal function (poor neural connectivity)
 - Altered mitochondrial oxidative metabolism, increased free radical, increased inflammatory response
 - Increased lactic acid



Giza, Christopher C., and David A. Hovda. "The New Neurometabolic Cascade of Concussion." *Neurosurgery*. Print.

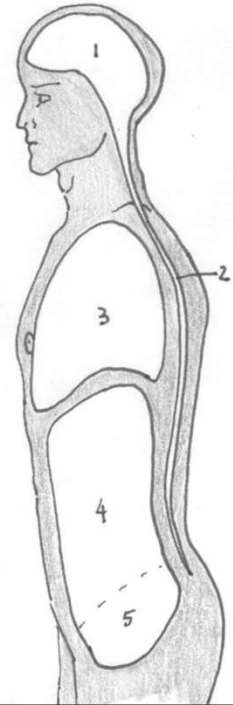


LYMPHATIC INFLUENCE

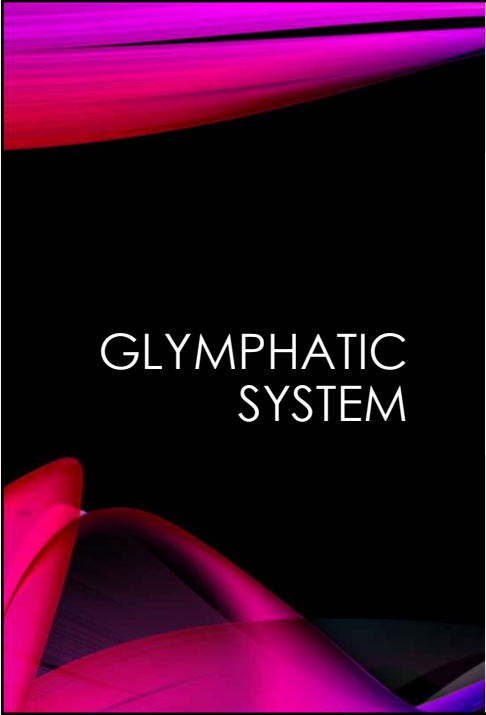
This inflammatory cascade could explain cognitive, mood, and sleep symptoms

Address the “baffles” of the body to help engage flow

- Tentorium cerebelli (and the rest of the cranial dura) [falx](#)
- Thoracic inlet
- Abdominal diaphragm
- Pelvic diaphragm



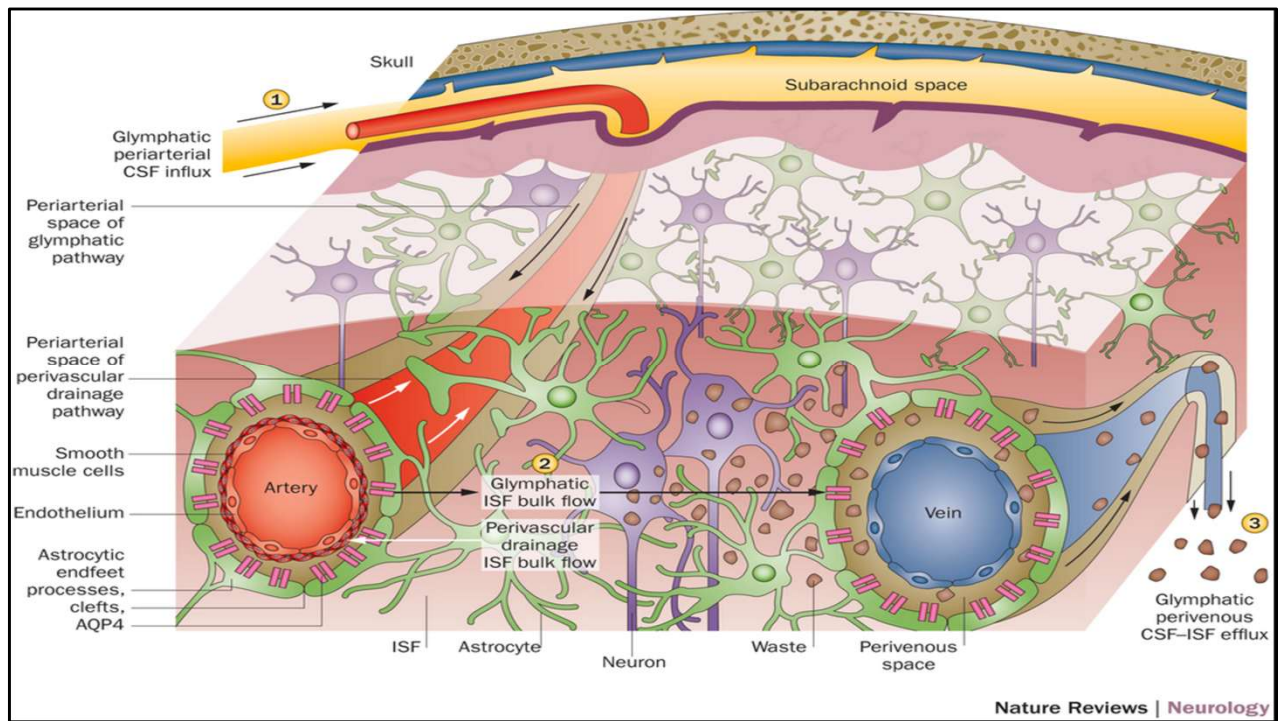
Respiratory circulatory model of osteopathy



GLYMPHATIC SYSTEM

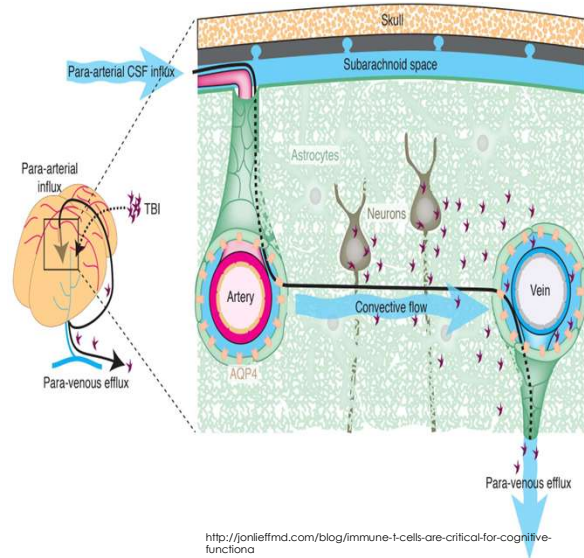
- Relatively newly discovered anatomy
- Waste disposal system for the CNS
- "Facilitates brain-wide distribution of several compounds, including glucose, lipids, amino acids, growth factors, and neuromodulators"
- Functions mainly during sleep
 - 60% increase in interstitial space and increased convective flux between CSF and interstitial fluid

altered due to trauma will get further accumulation of bypro
not sleeping well - system functioning even less



GLYMPHATIC STRUCTURE

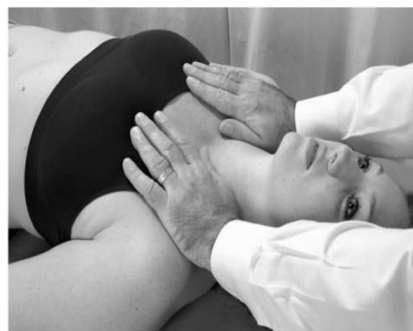
- Perivascular tunnel system (formed by astroglial cells)
- CSF $\longleftrightarrow$ Interstitial fluid
 - CSF driven into Virchow—Robin Space $\rightarrow$ interstitial space $\rightarrow$ perivenous spaces of large veins $\rightarrow$ deep cervical lymphatic system



Virchow-robin (space surrounding penetrating arterioles at pia/brain intervace)

LYMPHATIC TECHNIQUES AND TBI

- Performed on 24 comatose patients in ICU
 - 18-69 years old
- Treated with pedal and lymphatic pump
- Group 1: baseline ICP ≤ 20 mmHg: slight decrease in ICP, increased CCP
- Group 2: baseline ICP >20 mmHg: decreased ICP, increased CCP



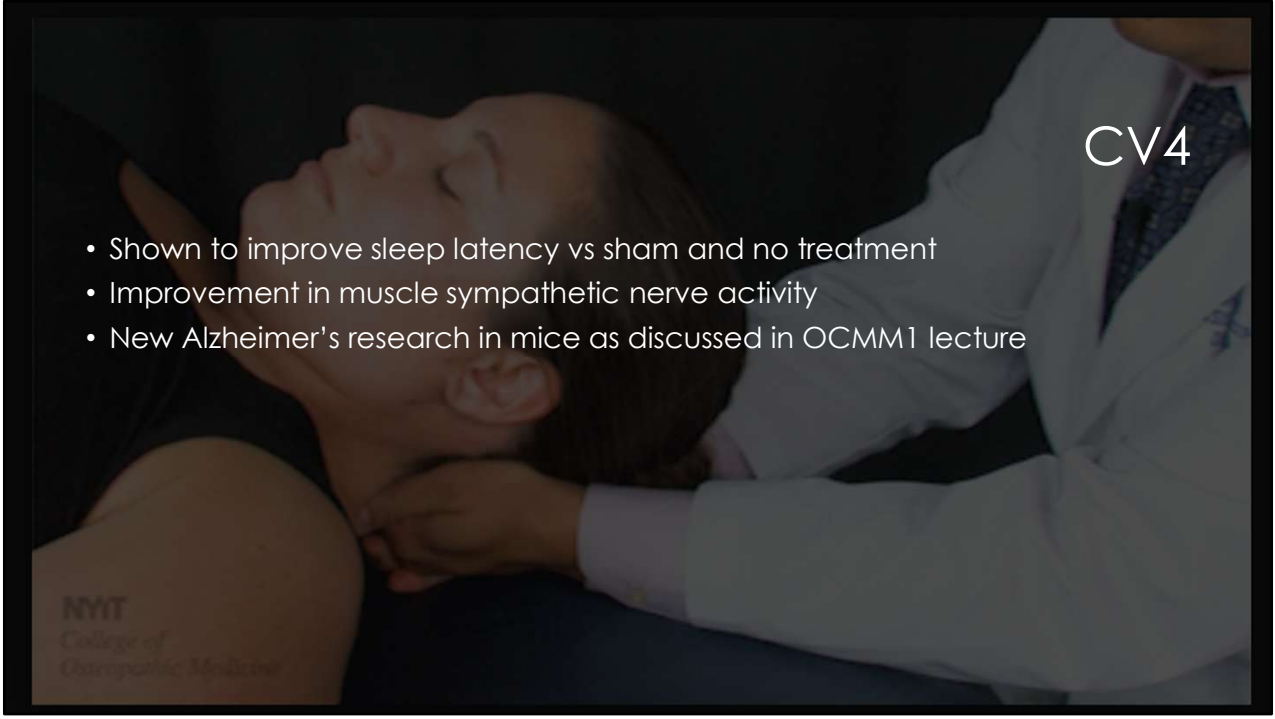
Yale.edu

Monitored at baseline, during OMT, and 5 minutes after
Drains open during OMT except each minute to register ICP
50 treatments with pedal/thoracic pump



LYMPHATIC TECHNIQUES

- With techniques that treat the lymphatic/fluid systems in the body studies have shown:
 - Increased flow toward the thoracic duct
 - Increased leukocyte release into lymph/mobilization of leukocytes
 - Mobilization of inflammatory modulators
 - Improved healing

A photograph of a person lying down with their eyes closed, receiving a neck massage from a practitioner wearing a white lab coat. The background is dark. The text 'CV4' is overlaid in the top right corner. A list of bullet points is overlaid in the center. In the bottom left corner, there is a logo for NYU College of Osteopathic Medicine.

CV4

- Shown to improve sleep latency vs sham and no treatment
- Improvement in muscle sympathetic nerve activity
- New Alzheimer's research in mice as discussed in OCMM1 lecture

NYU
College of
Osteopathic Medicine



PTSD AND HEAD INJURY

- Received "light touch manual therapy" – craniosacral therapy and brain curriculum
- Done on active-duty soldiers
- 10 men, 27-45 y/o
 - 80% TBI, 90% HA, 100% PTSD
- 2 one-hour sessions one week apart
- Showed decrease in intensity of HA and anxiety, increased cognitive function and mobility



CASE STUDIES

- 16 y/o F with 3 prior head injuries including most recent ~1 month prior
 - Sxs included amnesia, fatigue, HA, labile mood, vertigo, photo and phonophobia, concentration problems, insomnia, and vision problems
 - Resolution of symptoms over ~1 month of OMT
- 27 y/o with TBI after snowboarding
 - Had dizziness, nausea, fogginess, tinnitus
 - Resolution of symptoms following 25 minutes of OMT and improvement in sensory organization test following OMT
- 38 y/o F with 3 months of post-concussion symptoms
 - Multi-disciplinary approach including OMM, assessing psychological factors, pharmacologic interventions, injections, PT



OSTEOPATHIC RESEARCH

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- Patel KG, Sabini RC. Safety of osteopathic cranial manipulative medicine as an adjunct to conventional postconcussion symptom management: a pilot study. *J Am Osteopath Assoc*. 2018;118(6):403-409. doi: 10.7556/jaoa.2018.061
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